

- C_p = specific heat of the gas,
 T_1 = temperature of gas before heat is transferred to it,
 T_2 = temperature of gas after heat is transferred to it.

This equation can then be rearranged for Q :

$$Q = \frac{H}{C_p(T_2 - T_1)}.$$

A voltage signal can be obtained by this method of measurement which can be utilized in data logging.

Control of gas flow is usually by needle valves. Often this method of control is not sufficient, and it is necessary to incorporate a self-acting flow-control valve. At a small scale, such valves as the 'flowstat' are available (G.A. Platon, Ltd, Wella Road, Basingstoke, Hampshire, U.K.). Fluctuations in pressure in a flow-measuring orifice cause a valve or piston pressing against a spring to gradually open or close so that the original, preselected flow rate is restored. In a gas 'flowstat', the orifice should be upstream when the gas supply is at a regulated pressure and downstream when the supply pressure fluctuates and the back pressure is constant. Valves operating by a similar mechanism are available for larger scale applications.

LIQUIDS

The flow of non-sterile liquids can be monitored by a number of techniques (Howe *et al.*, 1969), but measurement of flow rates of sterile liquids presents a number of problems which have to be overcome. On a laboratory scale flow rates may be measured manually using a sterile burette connected to the feed pipe and timing the exit of a measured volume. The possible use of rotameters has already been mentioned in the previous section. A more expensive method is to use an electrical flow transducer (Howe *et al.*, 1969) which can cope with particulate matter in suspension and measure a range of flow rates from very low to high ($50 \text{ cm}^3 \text{ min}^{-1}$ to $500,000 \text{ dm}^3 \text{ min}^{-1}$) with an accuracy of $\pm 1\%$. In this flowmeter (Fig. 8.2) there are two windings outside the tube, supplied with an alternating current to create a magnetic field. The voltage induced in the field is proportional to the relative velocity of the fluid and the magnetic field. The potential difference in the fluid can be measured by a pair of electrodes, and is directly proportional to the velocity of the fluid.

In batch and fed-batch culture fermenters, a cheaper alternative is to measure flow rates indirectly by load cells (see Weight section). The fermenter and all ancillary reservoirs are attached to load cells which monitor

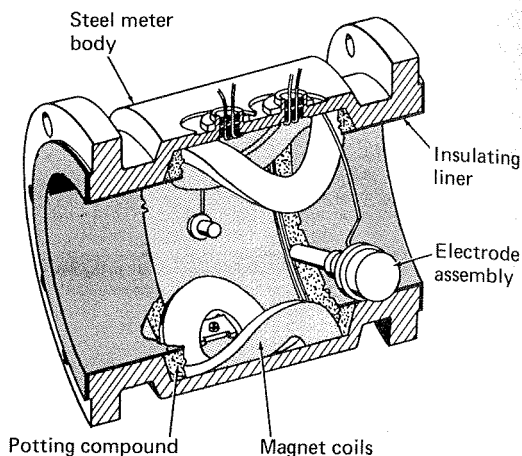


FIG. 8.2. A cut-away view of a short-form magnetic flowmeter (Howe *et al.*, 1969).

the increases and decreases in weight of the various vessels at regular time intervals. Provided the specific gravities of the liquids are known it is possible to estimate flow rates fairly accurately in different feed pipes. This is another technique which may be used with particulate suspensions.

Another indirect method of measuring flow rates aseptically is to use a metering pump which pumps liquid continuously at a predetermined and accurate rate. A variety of metering pumps are commercially available including motorized syringes, peristaltic pumps, piston pumps and diaphragm pumps. Motorized syringes are used only when very small quantities of liquid have to be added slowly to a vessel. In a peristaltic pump, liquid is moved forwards gradually by squeezing a tubing held in a semicircular housing. A variety of sizes of tubes can be used in different pumps to produce different known flow rates over a very wide range. Suspensions can be handled since the liquid has no direct contact with moving parts.

A piston pump contains an accurately machined ceramic or stainless-steel piston moving in a cylinder normally fitted with double ball inlet and outlet valves. The piston is driven by a constant-speed motor. Flow rates can be varied within a defined range by changing the stroke rate, the length of the piston stroke and by using a different piston size. Sizes are available from $\text{cm}^3 \text{ h}^{-1}$ to thousands of $\text{dm}^3 \text{ h}^{-1}$ and all can be operated at relatively high working pressures. Unfortunately, they cannot be used to pump fibrous or particulate suspensions. Piston pumps are more expensive than comparable sized peristaltic pumps but do not suffer from tube failure.

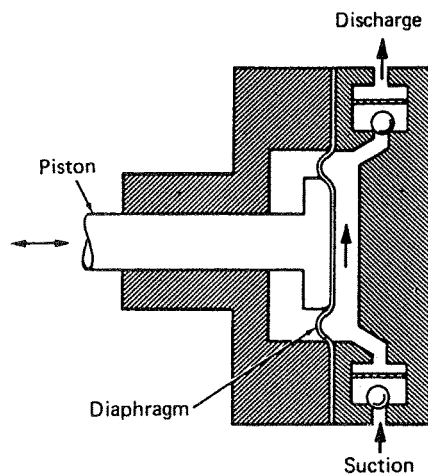


FIG. 8.3. A direct-driven diaphragm pump (Howe *et al.*, 1969).

Leakage can occur via the shaft housing of a piston pump. The problem can be prevented by the use of a diaphragm pump. This pump uses a flexible diaphragm to pump fluid through a housing (Fig. 8.3) with ball valves to control the direction of flow. The diaphragm may be made of, e.g., teflon, neoprene, stainless steel, and is actuated by a piston. A range of sizes of pumps is available for flow rates up to thousands of $\text{dm}^3 \text{h}^{-1}$.

Liquid flow from a nutrient feed tank or to or out of a fermenter may be monitored by continuous weighing on a balance or load cell(s). This will be discussed in the Weight section.

Pressure measurement

Pressure is one of the crucial measurements that must be made when operating many processes. Pressure measurements may be needed for several reasons, the most important of which is safety. Industrial and laboratory equipment is designed to withstand a specified working pressure plus a factor of safety. It is therefore important to fit the equipment with devices that will sense, indicate, record and control the pressure. The measurement of pressure is also important in media sterilization. In a fermenter, pressure will influence the solubility of gases and contribute to the maintenance of sterility when a positive pressure is present.

One of the standard pressure measuring sensors is the Bourdon tube pressure gauge (Fig. 8.4), which is used as a direct indicating gauge. The partial coil has an elliptical cross-section (A-A) which tends to be-

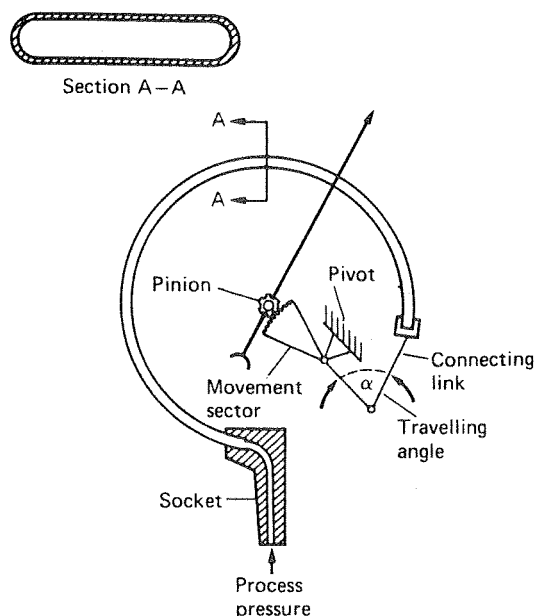


FIG. 8.4. 'C' Bourdon tube pressure gauge (Liptak, 1969).

come circular with increasing pressure, and because of the difference between the internal and external radii, gradually straightens out. The process pressure is connected to the fixed socket end of the tube while the sealed tip of the other end is connected by a geared sector and pinion movement which actuates an indicator pointer to show linear rotational response (Liptak, 1969).

When a vessel or pipe is to be operated under aseptic conditions a diaphragm gauge can be used (Fig. 8.5). Changes in pressure cause movements of the diaphragm capsule which are monitored by a mechanically levered pointer.

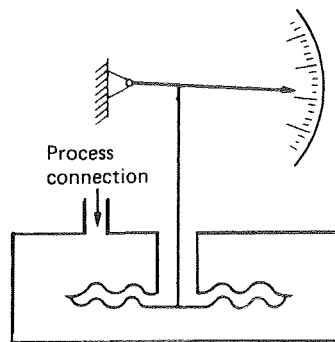


FIG. 8.5. Nested diaphragm-type pressure sensor (Liptak, 1969).

Alternatively, the pressure could be measured remotely using pressure bellows connected to the core of a variable transformer. The movement of the core generates a corresponding output. It is also possible to use pressure sensors incorporating strain gauges. If a wire is subject to strain its electrical resistance changes; this is due, in part, to the changed dimensions of the wire and the change in resistivity which occurs due to the stress in the wire. The output can then be measured over long distances. Another electrical method is to use a piezoelectric transducer. Certain solid crystals such as quartz have an asymmetrical electrical charge distribution. Any change in shape of the crystal produces equal, external, unlike electric charges on the opposite faces of the crystal. This is the piezoelectric effect. Pressure can therefore be measured by means of electrodes attached to the opposite surfaces of the crystal. Bioengineering AG (Wald, Switzerland) have made a piezoelectrical transducer with integral temperature compensation, to overcome pyroelectric effects, and built into a housing which can be put into a fermenter port.

It will also be necessary to monitor and record atmospheric pressure if oxygen concentrations in inlet and/or exit gases are to be determined using oxygen gas analysers (see later section). Paramagnetic gas analysers are susceptible to changes in barometric pressure. A change of 1% in pressure may cause a 1% change in oxygen concentration reading. This size of error may be very significant in a vessel where the oxygen consumption rate is very low and there is very little difference between the inlet and exit gas compositions. The pressure changes should be constantly monitored to enable the appropriate corrections to be made.

Pressure control

Different working pressures are required in different parts of a fermentation plant. During normal operation a positive head pressure of 1.2 atmospheres (161 kN^{-1}) absolute is maintained in a fermenter to assist in the maintenance of aseptic conditions. This pressure will obviously be raised during a steam-sterilization cycle (Chapter 5). The correct pressure in different components should be maintained by regulatory valves (Chapter 7) controlled by associated pressure gauges.

Safety valves

Safety valves (Chapter 7) should be incorporated at various suitable places in all vessels and pipe layouts

which are likely to be operated under pressure. The valve should be set to release the pressure as soon as it increases markedly above a specified working pressure. Other provisions will be necessary to meet any containment requirements.

Agitator shaft power

A variety of sensors can be used to measure the power consumption of a fermenter. On a large scale, a watt meter attached to the agitator motor will give a fairly good indication of power uptake. This measuring technique becomes less accurate as there is a decrease in scale to pilot scale and finally to laboratory fermenters, the main contributing factor being friction in the agitator shaft bearing (Chapter 7). Torsion dynamometers can be used in small-scale applications. Since the dynamometer has to be placed on the shaft outside the fermenter the measurement will once again include the friction in the bearings. For this reason strain gauges mounted on the shaft within the fermenter are the most accurate method of measurement and overcome frictional problems (Aiba *et al.*, 1965; Brodgesell, 1969). Aiba *et al.* (1965) mounted four identical strain gauges at 45° to the axis in a hollow shaft. Lead wires from the gauges passed out of the shaft via an axial hole and electrical signals were then picked up by an electrical slip-ring arrangement. Theoretical treatment of the strain gauge measurements has been covered by Aiba *et al.* (1973).

Rate of stirring

In all fermenters it is important to monitor the rate of rotation (rpm) of the stirrer shaft. The tachometer used for this purpose may employ electromagnetic induction voltage generation, light sensing or magnetic force as detection mechanisms (Brodgesell, 1969). Obviously, the final choice of tachometer will be determined by the type of signal which is required for recording and/or process control for regulating the motor speed and other ancillary equipment. Provision is often made on small laboratory fermenters to vary the rate of stirring. In most cases it is now standard practice to use an a.c. slip motor that has an acceptable torque curve that is coupled to a thyristor control. At pilot or full scale, the need to change rates of stirring is normally reduced. When necessary it can be done using gear boxes, modifying the sizes of wheels and drive belts, or by changing the drive motor, the most expensive alternative.

Foam sensing and control

The formation of foam is a difficulty in many types of microbial fermentation which can create serious problems if not controlled. It is common practice to add an antifoam to a fermenter when the culture starts foaming above a certain predetermined level. The methods used for foam sensing and antifoam additions will depend on process and economic considerations. The properties of antifoams have been discussed elsewhere (Chapters 4 and 7), as has their influence on dissolved oxygen concentrations (Chapter 9).

A foam sensing and control unit is shown in Fig. 8.6. A probe is inserted through the top plate of the fermenter. Normally the probe is a stainless-steel rod, which is insulated except at the tip, and set at a defined level above the broth surface. When the foam rises and touches the probe tip, a current is passed through the circuit of the probe, with the foam acting as an electrolyte and the vessel acting as an earth. The current actuates a pump or valve and antifoam is released into the fermenter for a few seconds. Process timers are routinely included in the circuit to ensure that the antifoam has time to mix into the medium and break down the foam before the probe is programmed after a preset time interval to sense the foam level again and possibly actuate the pump or valve. Alternatively antifoam may be added slowly at a predetermined rate by a small pump so that foaming never occurs and there is therefore no need for a sensing system.

A number of mechanical antifoam devices have been described including discs, propellers, brushes or hollow cones attached to the agitator shaft above the surface of the broth. The foam is broken down when it is thrown against the walls of the fermenter. Other de-

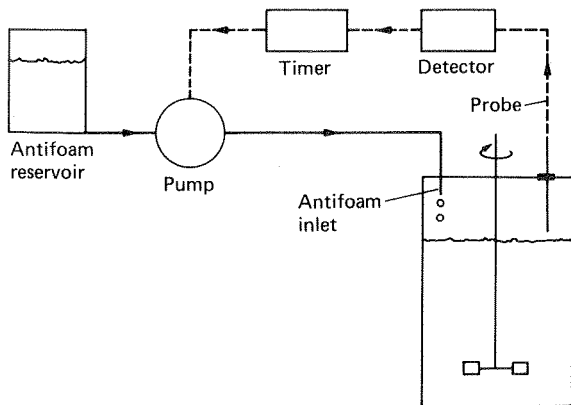


FIG. 8.6. Foam sensing and control unit.

vices which have been manufactured include horizontal rotating shafts, centrifugal separators and jets spraying on to deflector plates (Hall *et al.*, 1973; Viesturs *et al.*, 1982). Unfortunately most of these devices have to be used in conjunction with an antifoam.

Weight

A load cell offers a convenient method of determining the weight of a fermenter or feed vessel. This is done by placing compression load cells in or at the foot of the vessel supports. When designing the support system for a fermenter or other vessel, the weight of which is to be measured by load cells, the principle of the three-legged stool should be remembered. Three feet will always rest in stable equilibrium even though the supporting surface is uneven. If more feet are provided, the additional feet must each be fitted with means of adjustment or precision packing to ensure load bearing on all the feet.

A load cell is essentially an elastic body, usually a solid or tubular steel cylinder, the compressive strain of which under axial load may be measured by a series of electrical resistance strain gauges which are cemented to the surface of the cylinder. The load cell is assembled in a suitable housing with electrical cable connecting points. The cell is calibrated by measuring compressive strain over the appropriate range of loading. Changes of resistance with strain which are proportional to load are determined by appropriate electrical apparatus.

It is therefore possible to use appropriately sized load cells to monitor feed rates from medium reservoirs, acid and base utilization for pH control and the use of antifoam for foam control. The change in weight in a known time interval can be used indirectly as a measure of liquid flow rates.

Microbial biomass

Real-time estimation of microbial biomass in a fermenter is an obvious requirement, yet it has proved very difficult to develop a satisfactory sensor. Most monitoring has been done indirectly by dry weight samples (made quicker with microwave ovens), cell density (spectrophotometers), cell numbers (Coulter counters) or by the use of gateway sensors which will be discussed later in this chapter. Other alternative approaches are real-time estimation of a cell component which remains at a constant concentration, such

as nicotinamide adenine dinucleotide (NAD), by fluorimetry or measurement of a cell property which is proportional to the concentration of viable cells, such as radio frequency capacitance.

It is well established that fluorimetric measurements are very specific and rapid, but their use in fermentation studies is limited. The measurement of NAD, provided that it remains at a constant concentration in cells, would be an ideal indirect method for continuous measurement of microbial biomass. In pioneer studies, Harrison and Chance (1970) used a fluorescence technique to determine NAD–NADH levels inside microbial cells growing in continuous culture. Einsele *et al.* (1978) mounted a fluorimeter on a fermenter observation port located beneath the culture surface which enabled the measurement of NADH fluorescence *in situ*, making it possible to determine bulk mixing times in the broth and to follow glucose uptake by monitoring NADH levels. Beyeler *et al.* (1981) were able to develop a small sterilizable probe for fitting into a fermenter to monitor NADH, which had high specificity, high sensitivity, high stability and could be calibrated *in situ*. In batch culture of *Candida tropicalis*, the NAD(P)H-dependent fluorescence signal correlated well with biomass, so that it could be used for on-line estimation of biomass. Changes in the growth conditions, such as substrate exhaustion or the absence of oxygen, were also very quickly detected.

Schneckenburger *et al.* (1985) used this technique to study the growth of methanogenic bacteria in anaerobic fermentations. They thought cost was a problem, fluorescence equipment being too expensive for routine biotechnology applications when the minimum price was about US\$10,000. Ingold (Switzerland) have developed the Fluorosensor, a probe which can be integrated with a small computer or any data transformation device (Gary, Meier and Ludwig, 1988).

Dielectric spectroscopy can be used on-line to monitor biomass. Details of the theory and principles of this technique have been described by Kell (1987). At low radio frequencies (0.1 to 1.0 MHz), a microbial cell membrane will act as a capacitor, and become charged by the so-called β -dispersion effect (Schwan, 1957), making it possible to discriminate between microbial cells, gas bubbles and insoluble media particles. The size of this β -dispersion is linearly proportional to the membrane enclosed volume fraction up to high cell densities. Kell *et al.* (1987) were able to show that the capacitance (dielectric permittivity) was linearly proportional to the biomass concentration. The output for unicellular organisms is proportional to the mean cell radius whereas with mycelial suspensions the output

remains linear for increases in biomass of a particular cell morphology. The capacitance has been shown to give a linear response with biomass using a number of strains of bacteria, yeasts, mycelial fungi, plant and animal cells.

The sterilizable probe can be inserted directly into a fermenter using a 25-mm diameter port. Fouling of the gold electrodes in the probe can be avoided by the automatic application of electrolytic cleaning pulses. The sensor ('Bug meter') is manufactured by Aber Instruments (Aberystwyth, Wales) and marketed by Applikon (Schiedam, The Netherlands). It has a capacitance range of 0.1 to 200 pF (picoFarads), which is equivalent to 0.1 to 200 mg dry weight cm^{-3} (approximately 10^6 to 2×10^9 cells cm^{-3} of *Saccharomyces cerevisiae*). The resolution depends on the type of cells and the conductivity of the medium, but is normally 0.1 mg dry weight cm^{-3} . In order for the sensor to work effectively the suspending medium must have a minimum conductance. Yeast slurries after acid washing (Chapter 6) are satisfactory, but before such washing there may be a need for extra salts in the medium in order to make measurements. This sensor has proved ideal for yeast cells and is now being used by the brewing industry to control yeast pitching rates (Boulton *et al.*, 1989).

Measurement and control of dissolved oxygen

In most aerobic fermentations it is essential to ensure that the dissolved oxygen concentration does not fall below a specified minimal level. Since the 1970s steam sterilizable oxygen electrodes have become avail-

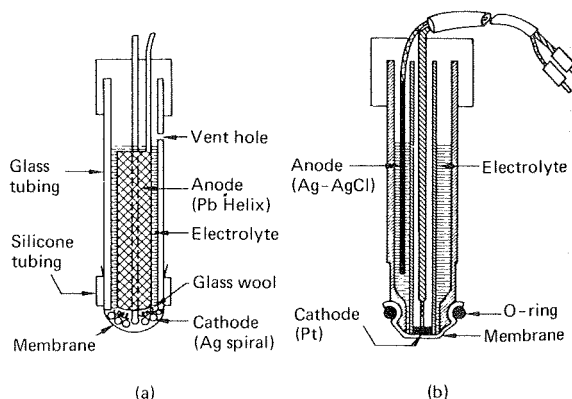


FIG. 8.7. Construction of dissolved-oxygen electrodes: (a) galvanic, (b) polarographic (Lee and Tsao, 1979).

able for this monitoring (Fig. 8.7). Details of electrodes are given by Lee and Tsao (1979).

These electrodes measure the partial pressure of the dissolved oxygen and not the dissolved oxygen concentration. Thus at equilibrium, the probe signal of an electrode will be determined by:

$$P(\text{O}_2) = C(\text{O}_2) \times P_T$$

where $P(\text{O}_2)$ is the partial pressure of dissolved oxygen sensed by the probe,

$C(\text{O}_2)$ is the volume or mole fraction of oxygen in the gas phase,

P_T is the total pressure.

The actual reading is normally expressed as percentage saturation with air at atmospheric pressure, so that 100% dissolved oxygen means a partial pressure of approximately 160 mmHg.

Pressure changes can have a significant effect on readings. If the total pressure of the gas equilibrating with the fermentation broth varies, the electrode reading will change even though there is no change in the gas composition. Changes in atmospheric pressure can often cause 5% changes and back pressure due to the exit filters can also cause increases in readings. Allowance must also be made for temperature. The output from an electrode increases by approximately 2.5% per °C at a given oxygen tension. This effect is due mainly to increases in permeability in the electrode membrane. Many electrodes have built-in temperature sensors which allow automatic compensation of the output signal. It is also important to remember that the solubility of oxygen in aqueous media is influenced by the composition. Thus, water at 25°C and 760 mmHg pressure saturated with air will contain 8.4 mg $\text{O}_2 \text{ dm}^{-3}$, while 25% NaCl in identical conditions will have an oxygen solubility of 2.0 mg $\text{O}_2 \text{ dm}^{-3}$. However, the measured partial pressure outputs for O_2 would be the same even though the oxygen concentrations would be very different. Therefore it is best to calibrate the electrode in percentage oxygen saturation. More details on oxygen electrodes and their calibration has been given by Halling (1990).

In small fermenters (1 dm^3), the commonest electrodes are galvanic and have a lead anode, silver cathode and employ potassium hydroxide, chloride, bicarbonate or acetate as an electrolyte. The sensing tip of the electrode is a teflon, polyethylene or polystyrene membrane which allows passage of the gas phase so that an equilibrium is established between the gas phases inside and outside the electrode. Because of the relatively slow movement of oxygen across the membrane, this type of electrode has a slow response of the

order of 60 seconds to achieve a 90% reading of true value (Johnson *et al.*, 1964). Buhler and Ingold (1976) quote 50 seconds for 98% response for a later version. These electrodes are therefore suitable for monitoring very slow changes in oxygen concentration and are normally chosen because of their compact size and relatively low cost. Unfortunately, this type of electrode is very sensitive to temperature fluctuations, which should be compensated for by using a thermistor circuit. The electrodes also have a limited life because of corrosion of the anode.

Polarographic electrodes, which are bulkier than galvanic electrodes, are more commonly used in pilot and production fermenters, needing instrument ports of 12, 19 or 25 mm diameter. Removable ones need a 25 mm port. They have silver anodes which are negatively polarized with respect to reference cathodes of platinum or gold, using aqueous potassium chloride as the electrolyte. Response times of 0.05 to 15 seconds to achieve a 95% reading have been reported (Lee and Tsao, 1979). The electrodes which can be very precise may be both pressure and temperature compensated. Although a polarographic electrode may initially cost 600% more than the galvanic equivalent, the maintenance costs are considerably lower as only the membrane should need replacing.

Prototypes of a fast response phase fluorometric sterilizable oxygen sensor are now being developed (Bambot *et al.*, 1994). The sensor utilizes the differential quenching of a fluorescence lifetime of a chromophore, tris(4,7-diphenyl-1,10-phenanthroline)ruthenium(II) complex, in response to the partial pressure of oxygen. The fluorescence of this complex is quenched by oxygen molecules resulting in a reduction of fluorescence lifetime. Thus, it is possible to obtain a correlation between fluorescence lifetime and the partial pressure of oxygen. However, at room temperature when a Clark-type oxygen electrode shows a linear calibration the optical sensor shows a hyperbolic response. The sensitivity of the optical sensor when compared with an oxygen electrode is significantly higher at low oxygen tensions whereas the sensitivity is low at high oxygen tensions. The sensor is autoclavable, free of maintenance requirements, stable over long periods and gives reliable measurements of low oxygen tensions in dense microbial cultures.

Dissolved oxygen concentrations may also be determined by a tubing method, described by Phillips and Johnson (1961) and Roberts and Shepherd (1968). The probe consists of a coil of permeable teflon or propylene tubing within the fermenter through which is

passed a stream of helium or nitrogen. The oxygen, which diffuses from the fermentation medium through the tubing wall into the inert gas stream, is then determined using a paramagnetic gas analyser (see next section). Times of 2 to 10 minutes are required before taking readings. The tubing will withstand repeated sterilization and has been used continuously for up to 1000 hours at pilot scale.

If it is necessary to increase the dissolved oxygen concentration in a medium this may be achieved by increasing the air flow rate or the rpm of the impeller or a combination of both processes. Another way is to increase the ratio of oxygen to nitrogen in the input gas using a variable proportionating valve while maintaining a constant gas flow rate (Siegal and Gaden, 1962). The cost of this technique would normally restrict its use to laboratories and pilot plants.

Inlet and exit-gas analysis

The measurement and recording of the inlet and/or exit gas composition is important in many fermentation studies. By observing the concentrations of carbon dioxide and oxygen in the entry and exit gases in the fermenter and knowing the gas flow rate it is possible to determine the oxygen uptake of the system, the carbon dioxide evolution rate and the respiration rate of the microbial culture.

The oxygen concentration can be determined by a paramagnetic gas analyser. Oxygen has a strong affinity for a magnetic field, a property which is shared with only nitrous and nitric oxides. The analysers may be of a deflection or thermal type (Brown *et al.*, 1969).

In the deflection analyser, the magnetic force acts on a dumb-bell test body that is free to rotate about an axis (Fig. 8.8). The magnetic force which is created around the test body is proportional to the oxygen concentration. When the test body swings out of the magnetic field a corrective electrostatic force must be applied to return it to the original position. Electrostatic force readings can therefore be used as a measure of oxygen concentrations.

In a thermal analyser, a flow through 'ring' element is the detector component (Fig. 8.9). After entering the ring, the paramagnetic oxygen content of the sample is attracted by the magnetic field to the central glass tube where resistors heat the gases. The resistors are connected into a Wheatstone bridge circuit to detect variations in resistance due to flow-rate changes. The oxygen in the heated sample loses a high proportion of its paramagnetism. Cool oxygen in the incoming gas flow

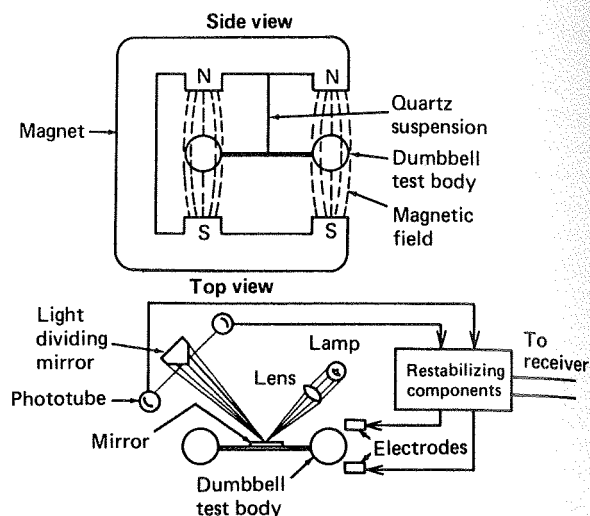


FIG. 8.8. A deflection-type paramagnetic oxygen analyser (Brown *et al.*, 1969).

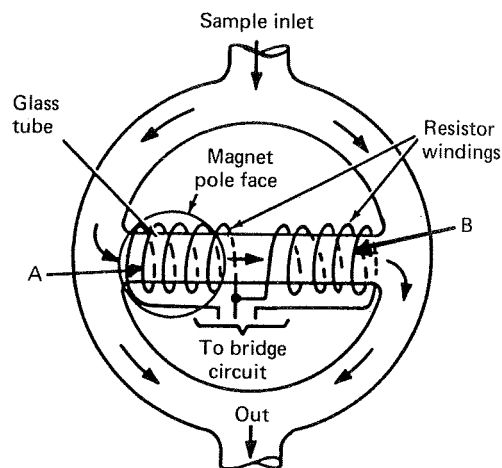


FIG. 8.9. The measuring element in a thermal-type paramagnetic oxygen analyser (Brown *et al.*, 1969).

will now be attracted and displace the hot oxygen. This displacement action produces a convection current. The flow rate of the convection current is a function of the oxygen concentration and can be detected by resistors. The resulting gas flow cools the winding A and heats the winding B, and it is the resulting temperature difference that imbalances the Wheatstone bridge.

Carbon dioxide is commonly monitored by infrared analysis using a positive filtering method. The unit (Fig. 8.10) consists of a source of infrared energy, a 'chopper'

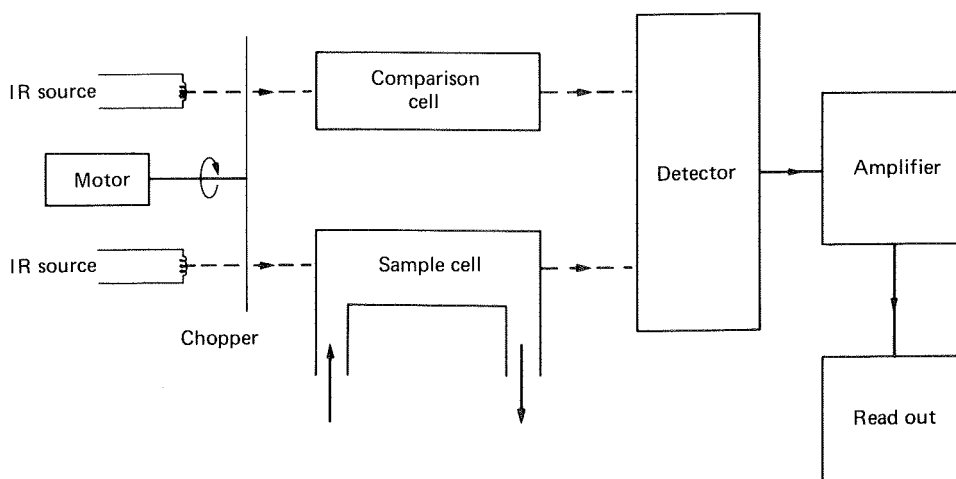


FIG. 8.10. Simple positive filtering infrared analyser.

to ensure that energy passes through each side of the optical system, a sample cell, a comparison (or reference) cell, and an infrared detector sensitized at a wavelength at which the gas of interest absorbs infrared energy. In this case the detector will be filled with carbon dioxide. This optical system senses the reduced radiation energy of the measuring beam reaching the detector, which is due to the absorption in the carbon dioxide in the sample cell.

It is expensive to have separate carbon dioxide and oxygen analysers for each separate fermenter. Therefore it may be possible to couple up a group of fermenters via a multiplexer to a single pair of gas analysers (Meiners, 1982). Gas analysis readings can then be taken in rotation for each fermenter every 30 to 60 minutes. In many cases this will be adequate. Alternatively the gas analysers can be replaced by a mass spectrometer which can analyse a number of components as well as oxygen and carbon dioxide (see later section).

pH measurement and control

In batch culture the pH of an actively growing culture will not remain constant for very long. In most processes there is a need for pH measurement and control during the fermentation if maximum yield of a product is to be obtained. Rapid changes in pH can often be reduced by the careful design of media, particularly in the choice of carbon and nitrogen sources, and also in the incorporation of buffers or by batch feeding (Chapters 2 and 4). The pH may be further

controlled by the addition of appropriate quantities of ammonia or sodium hydroxide if too acidic, or sulphuric acid if the change is to an alkaline condition. Normally the pH drift is only in one direction.

pH measurement is now routinely carried out using a combined glass reference electrode that will withstand repeated sterilization at temperatures of 121° and pressures of 138 kN m⁻². The electrodes may be silver/silver chloride with potassium chloride or special formulations (e.g. Friscoylt by Ingold) as an electrolyte. Occasionally calomel/mercury electrodes are used. The electrode is connected via leads to a pH meter/controller. If the electrode and its fermenter have to be sterilized in an autoclave then the associated leads and plugs to a pH meter must be able to withstand autoclaving and retain their electrical resistance. Repeated sterilization may gradually change the performance of the electrode. The long culture times associated with animal cell culture or continuous culture of any cells makes a withdrawal option highly desirable to allow for servicing the electrode or troubleshooting without interrupting the fermentation. The housing for this option needs to be carefully designed to ensure that the fermentation does not become contaminated when the electrode is withdrawn, serviced, resterilized and inserted into the housing (Gary *et al.*, 1988).

Ingold electrodes contain a ceramic housing in the reference half cell which has pore dimensions capable of preventing fungal or bacterial infections. However, it is often desirable in animal-cell culture to sterilize the reference electrolyte as well as the electrode surfaces and seals. Both liquid and gel filled electrodes are

available. The liquid system gives a faster response, is most stable and accurate. When the electrode is pressurized above the operating pressure in a fermenter the liquid electrolyte will gradually flow out of the ceramic diaphragm and prevent fouling, particularly from proteins in the fermentation broth precipitating on the membrane after contact with the electrolyte (Gary *et al.*, 1988).

Readers should consult Halling (1990) for more information on calibration and checking, sterilization, routine maintenance and problems in use.

Control units, to be discussed later in this chapter, may be simple ON/OFF or more complex. In the case of the ON/OFF controller, the controller is set to a predetermined pH value. When a signal actuates a relay, a pinch valve is opened or a pump started, and acid or alkali is pumped into the fermenter for a short time which is governed by a process timer (0 to 5 seconds). The addition cycle is followed by a mixing cycle which is governed by another process timer (0 to 60 seconds) during which time no further acid or alkali can be added. At the end of the mixing cycle another pH reading will indicate whether or not there has been adequate correction of the pH drift. In small volumes the likelihood of overshoot is minimal. A recording unit may be wired to the pH meter to monitor the pH pattern throughout a process cycle.

Shinsky (1973) has discussed pH control of batch processes using proportional and proportional plus derivative control (see later section) when overshoot of the set point is to be avoided. In the case of proportional action, the controller must be adjusted so that a valve on an acid feed-line is shut when the error is zero. However, overshoot is possible as there may be a delay in closing the valve once the set point is achieved. In some cases the overshoot cannot be corrected because of lack of alkali nor may it be desirable. Therefore to preclude an overshoot, the valve must be closed before the controlled variable reaches the set point. This may be done using proportional plus derivative control. The derivative action will need careful adjustment. Too little derivative action will cause some overshoot while too much will lead to the premature closure of the valve. This premature closure may be only for a short time before the valve opens again to give a response pattern as shown in Fig. 8.11.

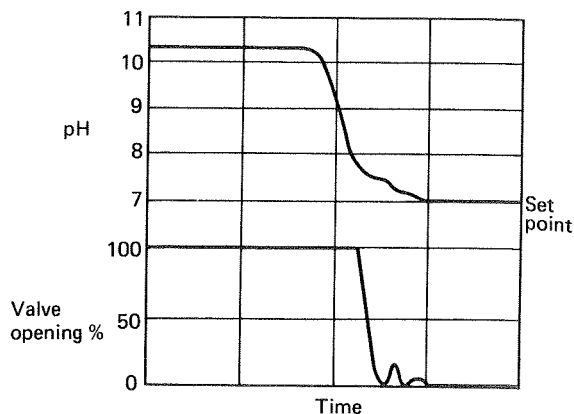
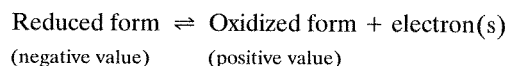


FIG. 8.11. Valve opening and pH changes with proportional plus derivative control (Shinsky, 1973).

is a measure of the oxidation-reduction potential of a biological system and can be determined as a voltage (mV), the value in any system depending on the equilibrium of:



The measuring electrode consists of gold, platinum or iridium which is welded to a copper lead. The interpretation of results presents difficulties and is confusing (Halling, 1990). The culture is not at redox equilibrium until possibly the end of a growth cycle. During the cycle, although some redox half-reactions may be in equilibrium they cannot all be in a dynamic system. The micro-organisms can also be at a different redox potential from the broth. It has been speculated that the probe signal is indicating something about the relative concentrations of uncertain and probably varying chemical species! This is far from the ideal for a sensor which should only be measuring a specific factor. If the broth contains traces of oxygen this will probably dominate the signal. It is the ability to detect low concentrations of oxygen in media (1 ppm) where redox electrodes may have a good application for determining oxygen availability in anaerobic or micro-aerophilic processes operated on a small scale (Kjaergaard and Joergensen, 1979).

Halling (1990) gives further details on routine handling, sterilization, testing and calibration.

Redox

Aspects of redox potential have been reviewed by Jacob (1970), Kjaergaard (1977) and Halling (1990). It

Carbon dioxide electrodes

The measurement of dissolved carbon dioxide is possible with an electrode, since a pH or voltage change

can be detected as the gas goes into solution. The first available electrode consisted of a combined pH electrode with a bicarbonate buffer (pH 5) surrounding the bulb and ceramic plug, with the solution being retained by a PTFE membrane held by an O-ring. Unfortunately, this electrode was not steam sterilizable. This basic design has been modified so that dissolved CO_2 from the sample permeates a stainless-steel reinforced silicone bi-layer membrane and dissolves in the internal bicarbonate electrolyte of the electrode. The subsequent pH shift is determined by an internal pH element. This system is extremely sensitive to shifts in the pH element as one pH unit change represents a tenfold increase in $p\text{CO}_2$, but precautions have been taken in the design to ensure minimal drift of the sensor. It is possible to calibrate the electrode on-line using specially formulated buffer solutions. This new version can be steam sterilized (Gary *et al.*, 1988).

ON-LINE ANALYSIS OF OTHER CHEMICAL FACTORS

If good control of a fermentation is to be obtained, then all chemical factors which can influence growth and product formation ought to be continuously monitored. This ideal situation has not yet been achieved but a number of techniques are currently being developed.

Ion-specific sensors

Ion-specific sensors have been developed by Orion Research and Radiometer to measure NH_4^+ , Ca^{2+} , K^+ , Mg^{2+} , PO_4^{3-} , SO_4^{2-} , etc. (Orion Research Inc., 11 Blackstone Street, Cambridge, MA, U.S.A.; Radiometer — Aagard Nielson and Schroder, Emdrupvej 72, Copenhagen-NV, Denmark). The response time of these electrodes varies from 10 seconds to several minutes depending on the concentration of the ion species, the composition of the sample, etc. However, none of these probes is steam sterilizable.

Enzyme and microbial electrodes

Enzyme or microbial cell electrodes can be used in some analyses. A suitable enzyme or microbial cell which produces a change in pH or forms oxygen in the enzyme reaction is chosen and immobilized on a membrane held in close contact to a pH or oxygen elec-

trode. Unfortunately, the oxygen demand of an enzyme may restrict the maximum substrate concentration which might be detected in a medium. Enfors (1981) overcame this problem for glucose determination by co-immobilizing glucose oxidase and catalase. More recently, it has been possible to use a ferrocene derivative as an artificial redox carrier to shuttle electrons from glucose oxidase to a carbon electrode, thus making the device largely independent of oxygen concentrations (Higgins and Boldot, 1992). Enzyme electrodes are also commercially available to monitor cholesterol, triglycerides, lactate, acetate, oxalate, methanol, ethanol, creatine, ammonia, urea, amino acids, carbohydrates and penicillin (Higgins *et al.*, 1987; Luong *et al.*, 1988; Higgins and Boldot, 1992).

Ideally, an electrode that can be inserted into a fermenter and steam sterilized is required, but none is yet available which can be used in this manner, even when using enzymes which are stable at high temperatures which can be obtained from thermophilic microorganisms. Hewetson *et al.* (1979) prepared sterilized penicillinase electrodes by assembling sterile components or by standing assembled components in chloroform before placing in a fermenter.

Near infra-red spectroscopy

Hammond and Brookes (1992) have described the development of near infra-red spectroscopy (NIR; 460–1200 nm) for rapid, continuous and batch analysis of components of fermentation broths. In samples from an antibiotic fermentation, they used NIR absorbance bands to simultaneously estimate fat (in the medium), teichoic acid (biomass) and antibiotic (the product). Fat analysis has been made possible with a fibre optic sensor placed *in situ* through a port in the fermenter wall. The assay time for an antibiotic has been reduced from 2 hours to 2 minutes. A method has also been developed to measure alkaline protease production in broths. Vaccari *et al.* (1994) have used this technique to measure glucose, lactic acid and biomass in a lactic acid fermentation.

Mass spectrometers

The mass spectrometer can be used for on-line analysis since it is very versatile and has a response time of less than 5 seconds for full-scale response and taking about 12 seconds for a sample stream. It allows for

monitoring of gas partial pressures (O_2 , CO_2 , CH_4 , etc.), dissolved gases (O_2 , CO_2 , CH_4 , etc.) and volatiles (methanol, ethanol, acetone, simple organic acids, etc.). Heinze *et al.* (1981) combined a data processor with a mass spectrometer equipped with a capillary gas inlet to measure gas partial pressures and a membrane inlet to detect dissolved gases and volatiles. It is therefore possible to multiplex this type of system to analyse several fermenters sequentially. Merck and Company Inc. have installed an integrated gas- and liquid-phase analysis system incorporating a mass spectrometer to sample a number of fermenters (Omstead and Greasham, 1989). The availability of low cost mass spectrometers at approximately \$15,000 does make their use in a laboratory or pilot plant financially feasible as an alternative to oxygen and carbon dioxide analysers, as well as analysing for a range of other gaseous and volatile compounds.

CONTROL SYSTEMS

The process parameters which are measured using probes described in the previous sections may be controlled using control loops.

A control loop consists of four basic components:

1. A measuring element.
2. A controller.
3. A final control element.
4. The process to be controlled.

In the simplest type of control loop, known as feedback control (Fig. 8.12), the measuring element senses a process property such as flow, pressure, temperature, etc., and generates a corresponding output signal. The controller compares the measurement signal with a predetermined desired value (set point) and produces an output signal to counteract any differences between the two. The final control element receives the control signal and adjusts the process by changing a valve opening or pump speed and causing the controlled process property to return to the set point.

Manual control

A simple example of control is manual control of a steam valve to regulate the temperature of water flowing through a pipe (Fig. 8.13). Throughout the time of operation a plant operative is instructed to monitor the temperature in the pipe. Immediately the tempera-

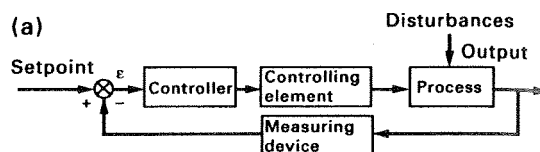


FIG. 8.12. A feedback control loop (Rolf and Lim, 1985).

ture changes from the set point on the thermometer, the operative will take appropriate action and adjust the steam valve to correct the temperature deviation. Should the temperature not return to the set point within a reasonable time, further action may be necessary. Much depends on the skill of individual operatives in knowing when and how much adjustment to make. This approach with manual control may be very costly in terms of labour and should always be kept to a strict minimum when automatic control could be used instead. A justifiable use of manual control may be in the adjustment of minor infrequent deviations.

Automatic control

When an automatic control loop is used, certain modifications are necessary. The measuring element must generate an output signal which can be monitored by an instrument. In the case of temperature control, the thermometer is replaced by a thermocouple, which is connected to a controller which in turn will produce a signal to operate the steam valve (Fig. 8.14).

Automatic control systems can be classified into four main types:

1. Two-position controllers (ON/OFF).
2. Proportional controllers.
3. Integral controllers.
4. Derivative controllers.

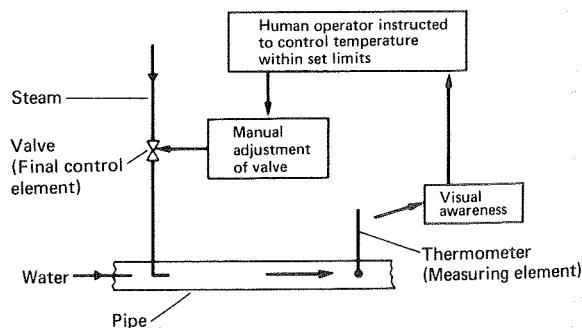


FIG. 8.13. Simple manual-control loop for temperature control.

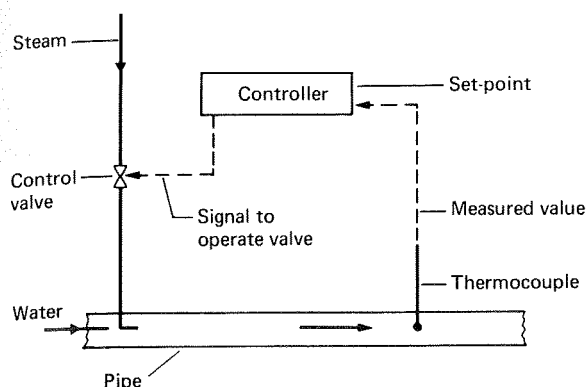


FIG. 8.14. Simple automatic control loop for temperature control.

TWO-POSITION CONTROLLERS (ON/OFF)

The two-position controller, which is the simplest automatic controller, has a final control unit (valve, switch, etc.) which is either fully open (ON) or fully closed (OFF). The response pattern to such a change will be oscillatory. If there is instant response then the pattern will be as shown in Fig. 8.15.

If one considers the example of the heating of a simple domestic water tank controlled by a thermostat operating with ON/OFF, then there will be a delay in response when the temperature reaches the set point and the temperature will continue rising above this point before the heating source is switched off. At the other extreme, the water will continue cooling after the heating source has been switched on. With this mode of operation an oscillatory pattern will be obtained with a repeating pattern of maximum and minimum temperature oscillating about the set point, provided that all the other process conditions are maintained at a steady level (Fig. 8.16).

If this type of controller is to be used in process control then it is important to establish that the maxi-

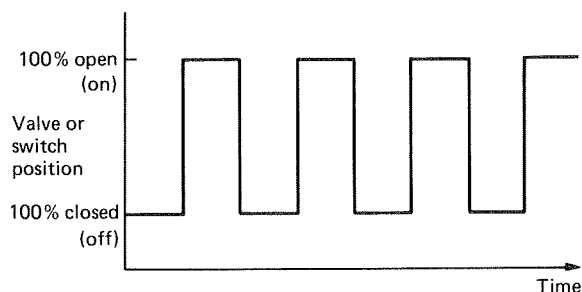


FIG. 8.15. Oscillatory pattern of a simple two-position valve or switch.

imum and minimum values are acceptable for the specific process, and to ensure that the oscillation cycle time does not cause excessive use of valves or switches.

ON/OFF control is not satisfactory for controlling any process parameter where there are likely to be large sudden changes from the equilibrium. In these cases alternative forms of automatic control must be used.

In more complex automatic control systems three different methods are commonly used in making error corrections. They are: proportional, integral and derivative. These control methods may be used singly or in combinations in applying automatic control to a process, depending upon the complexity of the process and the extent of control required. Since many of the controllers used in the chemical industries are pneumatic, the response to an error by the controller will be represented by a change in output pressure. Pneumatic controllers are still widely used because they are robust and reliable. In other cases, when the controller is electronic, the response to an error will be represented as a change in output current or voltage.

PROPORTIONAL CONTROL

Proportional control can be explained as follows: the change in output of the controller is proportional to the input signal produced by the environmental change (commonly referred to as error) which has been detected by a sensor.

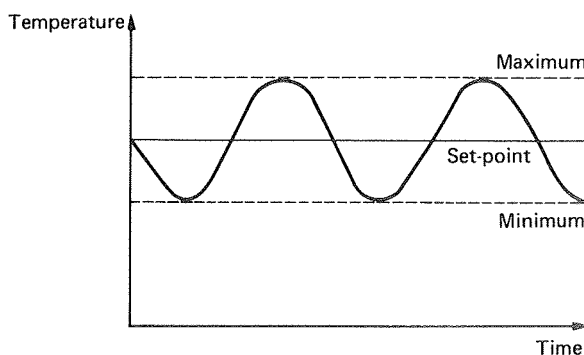


FIG. 8.16. Oscillatory pattern of the temperature of a domestic water tank (no water being drawn off) using ON/OFF control of the heating element.

- Output without control
- Proportional action
- Integral action
- Proportional + integral action
- Proportional + derivative action
- Proportional + integral + derivative action

Mathematically it can be expressed by the following equation:

$$M = M_0 + K_c \Sigma$$

where M = output signal,

M_0 = controller output signal when there is no

error,

K_c = controller gain or sensitivity,

Σ = the error signal.

Hence, the greater the error (environmental change) the larger is the initial corrective action which will be applied. The response to proportional control is shown in Fig. 8.17, from which it may be observed that there is a time of oscillation which is reduced fairly quickly. It should also be noted that the controlled variable attains a new equilibrium value. The difference between the original and the new equilibrium value is termed the offset.

The term K_c (controller gain) is the multiplying factor (which may be dimensioned) which relates a change in input to the change in output.

$$\Delta I \rightarrow \boxed{\text{Controller}} \rightarrow \Delta O$$

(Change in input) (Change in output)

Then

$$\Delta I = K_c \Delta O.$$

K_c may contain conversion units if there is an electrical input and a pressure output or vice versa.

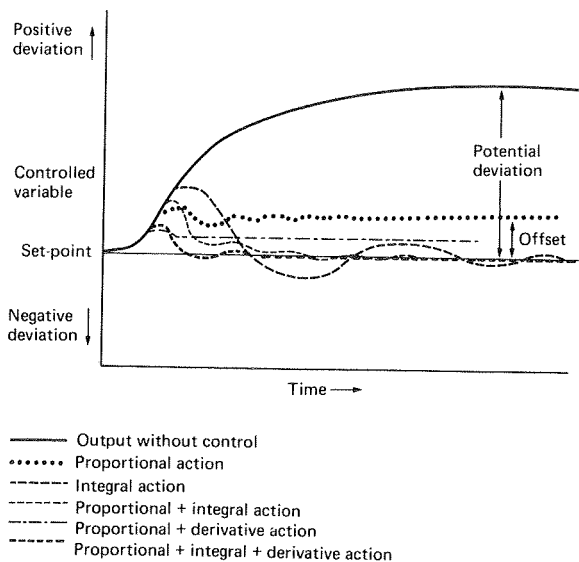


FIG. 8.17. Typical controlled plant responses.

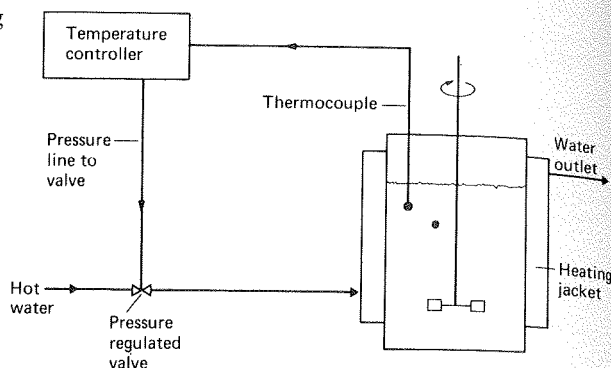


FIG. 8.18. A fermenter with a temperature-controlled heating jacket.

If the input to the controller is 1 unit of change, then:

- with a controller gain of 1, the output will be 1 unit,
- with a controller gain of 2, the output will be 2 units, etc.

On many controllers K_c is graduated in terms of proportional band instead of controller gain.

Now $K_c \propto \frac{1}{PB}$ (proportional band)

or $K_c = c' \frac{1}{PB}$ where c' is a constant.

This quantity PB is defined as the error required to move the final control element over the whole of its range (e.g. from fully open to fully shut) and is expressed as a percentage of the total range of the measured variable (e.g. two extremes of temperature).

A fermenter with a heating jacket will be used as an example (Fig. 8.18). A thermocouple is connected to a temperature controller which has a span of 10° covering the range 25° to 35° , with a set point at 30° . The controller valve which is controlled by a pressure regulator, is fully open at 5 psig (46 kN m^{-2}) and fully closed at 15 psig (138 kN m^{-2}), while the set point of 30° corresponds to a control pressure on the valve of 10 psig (92 kN m^{-2}). When the controller gain is 1, a change of 10° will cause a pressure change of 10 psig when the valve will be fully open at 25° and fully closed at 35° . Thus, the proportional band is 100%. If the controller gain is 2, a 5° change will cause the valve to go from fully open to fully closed, i.e. 27.5° to 32.5° . In this case the proportional band width will be:

$$\frac{\text{Actual band width}}{\text{Total band range}} = \frac{5^\circ}{10^\circ} \times 100 = 50\%.$$

Either side of this band the pressure will be constant. In the case of a controller gain of 4, a 2.5° change will cause a pressure change of 10 psig (92 kN m^{-2}), which will cause the control valve to go from fully open to fully closed (28.75° to 31.25°). In this case the proportional band will be

$$= \frac{2.5^\circ}{10^\circ} \times 100 = 25\%.$$

These results are summarized in Table 8.2 for controller gains of 1, 2 and 4.

When the proportional band is very small (the controller gain is high) the control mode can be likened to simple ON/OFF, with a high degree of oscillation but no offset. As the proportional band is increased (low controller gain) the oscillations are reduced but the offset is increased. Settings for proportional band width are normally a compromise between degree of oscillation and offset. If the offset is not desirable it can be eliminated by the use of proportional control in association with integral control (see Fig. 8.17 and later section).

INTEGRAL CONTROL

The output signal of an integral controller is determined by the integral of the error input over the time of operation. Thus:

$$M = M_0 + \frac{1}{T_i} \int \Sigma dt$$

where T_i = integral time.

It is important to remember that the controller

output signal changes relatively slowly at first as time is required for the controller action to integrate the error. It is evident from Fig. 8.17 that the maximum deviation from the set point is significant when compared with the use of proportional control for control of the chosen parameter, and the system takes longer to settle down. There is, however, no offset which is advantageous in many control processes.

DERIVATIVE CONTROL

When derivative control is applied the controller senses the rate of change of the error signal and contributes a component of the output signal that is proportional to a derivative of the error signal. Thus:

$$M = M_0 + T_d \frac{d\Sigma}{dt}$$

where T_d is a time rate constant.

It is important to remember that if the error is constant there is no corrective action with derivative control. In practice, derivative control is never used on its own. The response curve has therefore been deliberately omitted from Fig. 8.17.

Figure 8.19 demonstrates the response of derivative control to sinusoidal error inputs. The output is always in a direction to oppose changes in error, both away from and towards the set point, which in this example results in a 90° phase shift. This opposition to a change has a fast damping effect and this property is very useful in combination with other modes of control which will be discussed later.

TABLE 8.2. The effect of controller gain (K_c) on band width of a proportional temperature controller

	Measured temperature	Pressure output (psig) at		
		$K_c = 1 \text{ psig}/1^\circ$	$K_c = 2 \text{ psig}/1^\circ$	$K_c = 4 \text{ psig}/1^\circ$
Set → point	35°	15	15	15
	34°	14	15	15
	33°	13	15	15
	32.5°	12.5	15	15
	32°	12	14	15
	31.25°	11.25	12.5	15
	31°	11	12	14
	30°	10	10	10
	29°	9	8	6
	28.75°	8.75	7.5	5
	28°	8	6	5
	27.5°	7.5	5	5
	27°	7	5	5
	26°	6	5	5
	25°	5	5	5
		Prop. band 100%	Prop. band 50%	Prop. band 25%

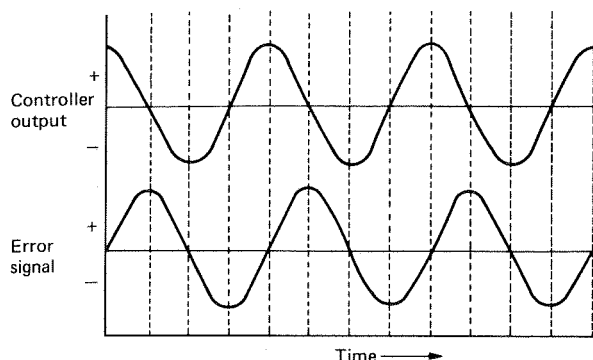


FIG. 8.19. Response of a derivative controller to sinusoidal error inputs.

Combinations of methods of control

Three combinations of control systems are used in practice:

- (a) Proportional plus integral.
- (b) Proportional plus derivative.
- (c) Proportional plus integral plus derivative.

PROPORTIONAL PLUS INTEGRAL CONTROL

When proportional plus integral control is used, the output response to an error gives rise to a slightly higher initial deviation in the output signal compared with that which would be obtained with proportional control on its own (Fig. 8.17). This is due to a contribution in the signal from integral control. However, the oscillations are soon reduced and there is finally no offset. This mode of control finds wide applications since the proportional component is ideal in a process where there are moderate changes, whereas the integral component will allow for large load changes and eliminate the offset that would have occurred.

PROPORTIONAL PLUS DERIVATIVE CONTROL

If proportional plus derivative control is used, the output response to an error will lead to reduced deviations, faster stabilization and a reduced offset (Fig. 8.17) compared with proportional control alone. Because the derivative component has a rapid stabilizing influence, the controller can cope with rapid load changes.

PROPORTIONAL PLUS INTEGRAL PLUS DERIVATIVE CONTROL

The combination of proportional plus integral plus derivative normally provides the best control possibilities (Fig. 8.17). The advantages of each system are retained. The maximum deviation and settling time are similar to those for a proportional plus derivative controller whilst the integral action ensures that there is no offset. This method of control finds the widest applications because of its ability to cope with wide variations of patterns of changes which might be encountered in different processes.

Controllers

The first primary automatic controllers were electronic control units which were adjusted manually to set up desired PID response patterns to a disturbance in a control loop. These controllers are relatively expensive and some knowledge of control engineering is necessary to make the correct adjustments to obtain the required control responses.

The availability of cheap computers and suitable computer programs to mimic PID control and handle a number of control loops simultaneously has made it possible to use some very complex control techniques for process optimization. Some aspects of this work on computer applications will be discussed later in this chapter.

More complex control systems

In certain situations PID control is not adequate to control a disturbance. Control may be difficult when there is a long time lag between a change in a manipulated variable and its effect on the measured variable. Consider the example of a heat exchanger where the water temperature is regulated by the flow rate of steam from a steam valve (Hall *et al.*, 1974). By the time the effects of a change in steam flow influence the hot water temperature, a considerable energy change has occurred in the heat exchanger which will continue to drive the hot water temperature away from the set-point after a correction has been made to the steam-flow valve. This lag will lead to cycling of the measured variable about the set point. Cascade control can solve this problem. When cascade control is used (Fig. 8.20), the output of one controller is the set-point for another. Each controller has its own measured variable with only the primary controller having an

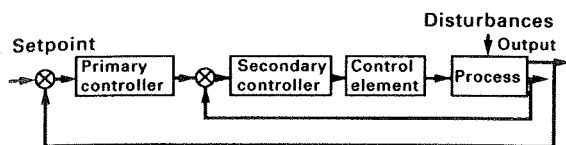


FIG. 8.20. A cascade feedback control loop (Rolf and Lim, 1985).

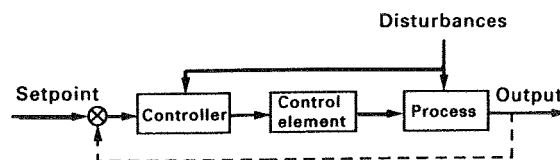
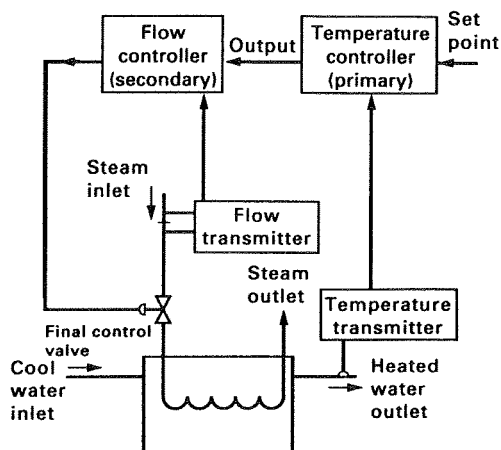
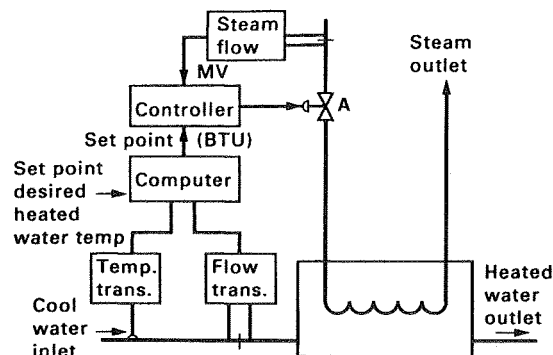


FIG. 8.22. A feedforward control loop (Rolf and Lim, 1985).

FIG. 8.21. Cascade control of a heat-exchange process (Hall *et al.*, 1974).FIG. 8.23. Feedforward control of a heat-exchange process (Hall *et al.*, 1974).

individual set-point and only the secondary controller providing an output to the process. In more complex cascade systems more loops may be included.

When cascade control is used to control water temperature in a heat exchanger process (Fig. 8.21) there is the primary loop (slow or outer loop), consisting of the temperature sensor and the primary temperature controller of the process water temperature, and the secondary loop (fast or inner loop) consisting of the steam flow sensor, the steam flow controller, its process variable (steam flow), the control valve and the process. The addition of the secondary controller, whose measured variable is steam flow, allows steam flow variations to be corrected immediately before they can affect the hot water temperature. This extra control loop should help to minimize temperature cycling.

Feed-forward control (Fig. 8.22) makes it possible to utilize other disturbances besides the measured values of the process that have to be controlled and enable fast control of a process. The cascade control system (Fig. 8.21) which has just been discussed would be adequate provided that the water inlet pressure and temperature do not fluctuate. When fluctuations occur, feed-forward control may be used, but this will require the measurement of both inlet water temperature and flow rate with both variables combined to control the

steam flow rate (Fig. 8.23). In the example shown (Hall *et al.*, 1974) a computer is programmed to determine the heat energy required to be added to the cool inlet water to bring it to the appropriate temperature at the outlet and allow the correct amount of steam to enter the heating coils. Inputs for temperature and flow are made from the inlet pipe to the computer. Calculations can then be made to determine the required heat input to raise the water temperature in the heat exchanger. The output control signal to the valve A is computed to allow the correct amount of steam to enter the heating coils.

When one or more of the process variables or characteristics is not known and cannot be measured directly, then adaptive control should be considered. The on-line identification of process characteristics and the subsequent use of this information to improve the process constitute adaptive control (Hall *et al.*, 1974). The sequences and the interactions in the adaptive control loop are outlined in Fig. 8.24. Adaptive control is useful in circumstances where the process dynamics are not well defined or change with time. This may be most useful in controlling a batch fermentation where considerable and often complex changes may occur (Bull, 1985; Dusseljee and Feijen, 1990). Adaptive control strategies have been used in fed-batch yeast cultivation (Montague *et al.*, 1988), amino acid production (Radjai *et al.*, 1984) and penicillin production (Lorenz *et al.*, 1985).

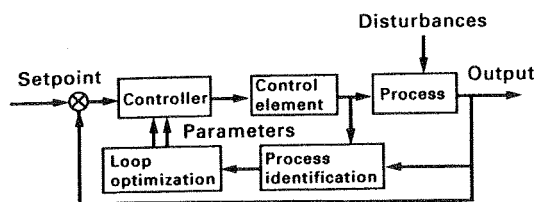


FIG. 8.24. Adaptive control (Rolf and Lim, 1985).

It is also possible to prepare sequential programs using a computer based control system. Consider control of the dissolved oxygen concentration in a fermentation broth. This may be changed by altering the agitation rate, the air or gas flow rate, the partial pressure of oxygen in the inlet gas or the total fermenter pressure. In practice, combinations of these variables may be used either sequentially or simultaneously using suitable computer programs. Initially the agitation rate can be increased to respond to decreases in dissolved oxygen concentration. When a predetermined maximum agitation rate is reached, the air flow can be steadily increased to a preset maximum, followed by the third and subsequent stages.

COMPUTER APPLICATIONS IN FERMENTATION TECHNOLOGY

Since the initial use of computers in the 1960s for modelling fermentation processes (Yamashita and Murao, 1978) and in process control for production of glutamic acid (Yamashita *et al.*, 1969) and penicillin (Grayson, 1969), there have been numerous publications on computer applications in fermentation technology (Rolf and Lim, 1985; Bushell, 1988; Whiteside and Morgan, 1989; Fish *et al.*, 1990). Initially, the use of large computers was restricted because of their cost but reductions in costs and the availability of cheaper small computers has widened interest in their possible applications. The availability of efficient small computers has led to their use for pilot plants and laboratory systems since the financial investment for the on-line computer amounts to a relatively insignificant part of the whole system.

Three distinct areas of computer function were recognized by Nyiri (1972):

1. *Logging of process data.* Data logging is performed by the data acquisition system which has both hardware and software components. There is an interface between the sensors and the computer. The software should include the com-

puter program for sequential scanning of the sensor signals and the procedure of data storage.

2. *Data analysis (Reduction of logged data).* Data reduction is performed by the data-analysis system, which is a computer program based on a series of selected mathematical equations. The analysed information may then be put on a print out, fed into a data bank or utilized for process control.
3. *Process control.* Process control is also performed using a computer program. Signals from the computer are fed to pumps, valves or switches via the interface. In addition the computer program may contain instructions to display devices or teletypes, to indicate alarms, etc.

At this point it is necessary to be aware that there are two distinct fundamental approaches to computer control of fermenters. The first is when the fermenter is under the direct control of the computer software. This is termed Direct Digital Control (DDC) and will be discussed in the next section. The second approach involves the use of independent controllers to manage all control functions of a fermenter and the computer communicates with the controller only to exchange information. This is termed Supervisory Set-Point Control (SSC) and will be discussed in more detail in the Process Control section.

It is possible to analyse data, compare it with model systems in a data store, and use control programs which will lead to process optimization. However, process optimization by this method is not a widely used procedure in the fermentation industries at present. It is important to be aware of these different applications, since this will influence the size and type of computer system which will be appropriate for the precise role that it is intended to perform, whether in a laboratory, a pilot plant, or manufacturing plant, or a combination of these three.

Components of a computer-linked system

When a computer is linked to a fermenter to operate as a control and recording system, a number of factors must be considered to ensure that all the components interact and function satisfactorily for control and data logging. A DDC system will be used as an example to explain computer controlled addition of a liquid from a reservoir to a fermenter. A simple outline of the main components is given in Fig. 8.25. A sensor S in the fermenter produces a signal which may need to

be amplified and conditioned in the correct analogue form. At this stage it is necessary to convert the signal to a digital form which can be subsequently transmitted to the computer. An interface is placed in the circuit at this point. This interface serves as the junction point for the inputs from the fermenter sensors to the computer and the output signals from the computer to the fermenter controls such as a pump T attached to an additive reservoir. Digital to analogue conversion is necessary between the interface and the pump T.

A sensor will generate a small voltage proportional to the parameter it is measuring. For example, a temperature probe might generate 1 V at 10°C and 5 V at 50°C. Unfortunately, the signal cannot be understood by the computer and must be converted by an analogue to digital converter (ADC) into a digital form.

The accuracy of an ADC will depend on the number of bits (the unit of binary information) it sends to the

computer. An 8-bit converter will work in the range 0–255 and it is therefore able to divide a signal voltage into 256 steps. This will give a maximum accuracy of $100/256$, which is approximately 0.4%. However, a 10-bit converter can give 1024 steps with an accuracy of $100/1024$, which is approximately 0.1%. Therefore when a parameter is to be monitored very accurately a converter of the appropriate degree of accuracy will be required. The time taken for an ADC to convert voltage signals to a digital output will vary with accuracy, but improved accuracy leads to slower conversion and hence slower control responses. However, cycle times of about 1 second may be adequate in many fermentation systems. It is also important to ensure that the voltage ranges of the sensors are matched to the ADC input range. More detailed discussion is given by Whiteside and Morgan (1989).

A digital to analogue converter (DAC) converts a

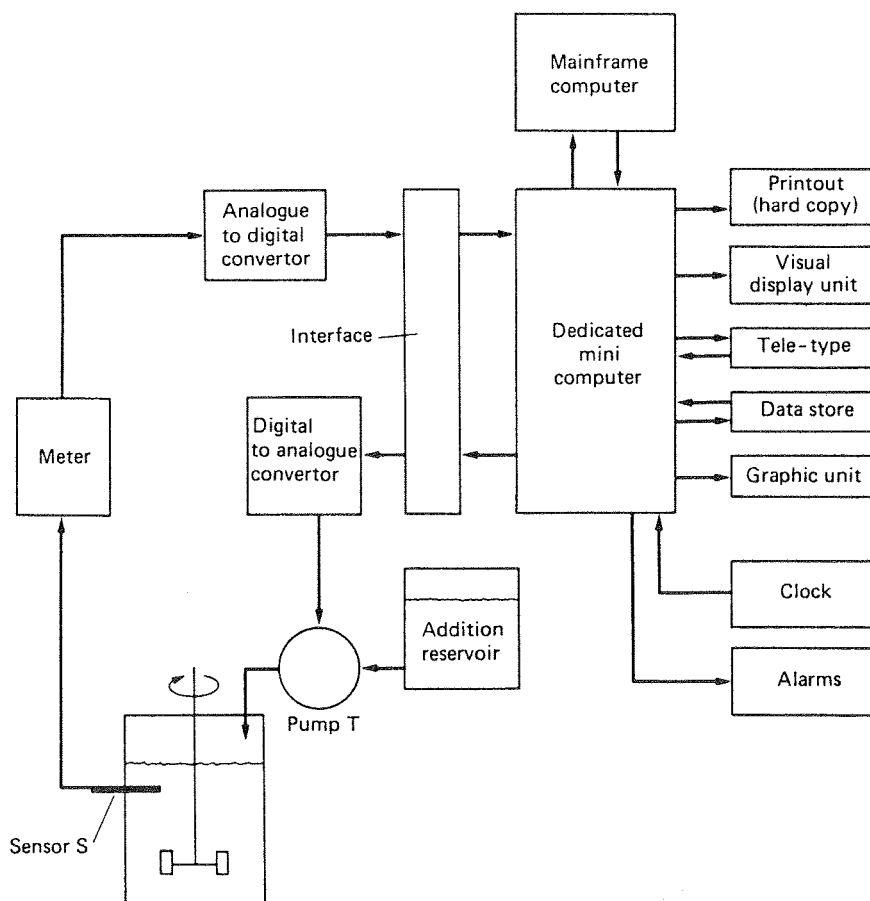


FIG. 8.25. Simplified layout of computer-controlled fermenter with only one control loop shown.

digital signal from the computer into an electrical voltage which can be used to drive electrical equipment, e.g. a stirrer motor. Like the ADC, the accuracy of the DAC will be determined by whether it is 8-bit, 10-bit, 12-bit, etc., and will for example determine the size of steps in the control of rpm of a stirrer motor.

The small computer itself is dedicated solely to one or more fermenters. This computer is coupled to a real-time clock, which determines how frequently readings from the sensor(s) should be taken and possibly recorded. The other ancillary equipment linked directly to the computer might include a visual display unit, a data store, a teletype, a graphic display unit, a print out, alarms and a barometer.

The small computer is often connected to a large main frame computer for random access, not on a real-time scale, but for long-term data storage and retrieval and for complex data analysis which will not be utilized subsequently in real-time control.

It is also possible to develop programs so that on-line instruments can be checked regularly and recalibrated when necessary. Swartz and Cooney (1979) were able to routinely recalibrate a paramagnetic oxygen analyser and an infrared carbon dioxide analyser every 12 hours utilizing a program which connected a gas of known composition to the analysers and subsequently monitored the analyser outputs.

Data logging

The simplest task for a computer is data logging. Parameters such as those listed in Table 8.1 can be measured by sensors which produce a signal which is compatible with the computer system.

Programs have been developed so that by reference to the real-time clock, the signals from the appropriate sensors will be scanned sequentially in a predetermined pattern and logged in a data store. Typically, this may be 2- to 60-second intervals, and the data is printed out on a visual display unit. In preliminary scanning cycles the values are compared with predefined limit values, and deviations from these values result in an error print out, or if more extreme then an alarm may be activated. In the final cycle of a sequence, say every 5 to 60 minutes, the program instructs that the sensor readings are permanently recorded on a print out or in a data store.

At the same time as on-line data is being recorded from sensors, analytical data for broth viscosity, microbial growth, substrate and precursor utilization and product formation, which have to be determined sepa-

rately may be logged into the data store for specific known times. Carleysmith (1989) has described a data handling system being used by Smith Kline Beecham in the United Kingdom.

Thus, it is now possible to record data continuously for a range of parameters from a number of fermenters simultaneously using minimal manpower, provided that the capital outlay is made for fermenters with suitable instrumentation coupled with adequate computer facilities.

Data analysis

Because a computer can undertake so many calculations very rapidly, it is possible to design programs to analyse fermentation data in a number of ways. A linked main-frame computer may be used for part of this analysis as well as the dedicated small computer.

A number of the monitoring systems were described as 'Gateway Sensors' by Aiba *et al.* (1973) and are given in Table 8.3. Gateway sensors are so called because the information they yield can be processed to give further information about the fermentation. More details of analysis of direct measurements, indirect measurements and estimated variables have been discussed by Zabriskie (1985) and Royce (1993).

The respiratory quotient of a culture may be calculated from the metered gas-flow rates and analyses for oxygen and carbon dioxide leaving a known volume of culture in the fermenter. This procedure was used to monitor growth of *Candida utilis* in a 250-dm³ fermenter, to follow or forecast events during operation (Nyiri *et al.*, 1975).

If one defines the fraction of substrate which is converted to product then it is possible to write mass

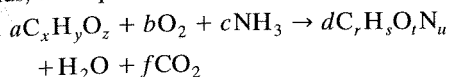
TABLE 8.3. Gateway sensors (Aiba *et al.*, 1973)

Sensor	Information that may be determined from the sensor signal
pH	Acid product formation
Dissolved oxygen	Oxygen-transferrate
Oxygen in exit gas	Oxygen-uptake rate
Gas-flow rate	
Carbon dioxide in exit gas	Carbon dioxide evolution rate
Gas-flow rate	
Oxygen-uptake rate	Respiratory quotient
Carbon dioxide evolution rate	
Sugar-level and feed rate	Yield and cell density
Carbon dioxide evolution rate	

balances for C, H, O and N with the measurement of only a few quantities (O_2 , CO_2 , NH_3 , etc.). All the other quantities can be calculated, including biomass and yield, if the biomass elemental composition is known (Chapter 2). This procedure was used for the analysis of a bakers' yeast fermentation (Cooney *et al.*, 1977). Biomass production can be regarded as a stoichiometric relationship in which substrate is converted, in the presence of oxygen and ammonia to biomass, carbon dioxide and water:

Carbon source-energy + oxygen + ammonium $\rightarrow$
cells + water + carbon dioxide.

Thus, the equation can be written in the form:



where a , b , c , d , e and f are moles of the respective reactants and products. $C_xH_yO_z$ is the molecular formula of the substrate where x , y and z are the specific carbon, hydrogen and oxygen atom numbers. Biomass is represented by $C_rH_sO_tN_u$ where r , s , t and u are the corresponding numbers of each element in the cell. This technique was developed to use with bakers' yeast fermentations (Cooney *et al.*, 1977; Wang *et al.*, 1977).

Process control

Arminger and Moran (1979) recognized three levels of process control that might be incorporated into a system. Each higher level involves more complex programs and needs a greater overall understanding of the

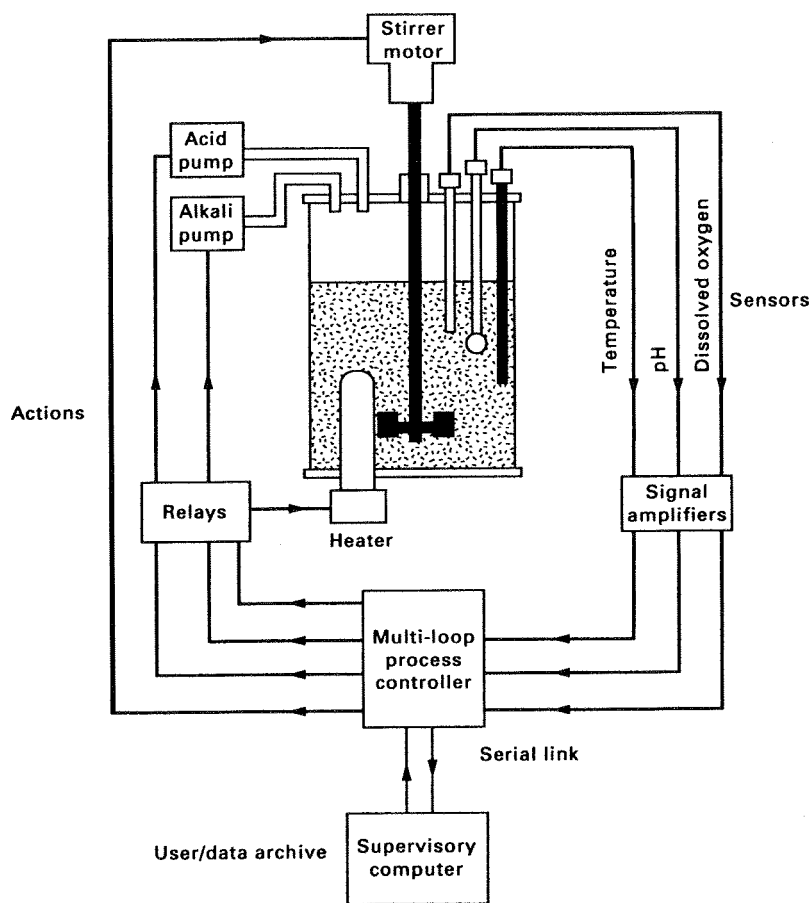


FIG. 8.26. Diagrammatic representation of a supervisory setpoint control (SSC) system for fermenters. This example illustrates a system controlling temperature by means of heating only, dissolved oxygen tension by stirrer speed and pH by the addition of acid and alkali. All control functions are performed by the intelligent process controller and the computer only communicates with this in order to log data and send new setpoints when instructed to do so by the user (Whiteside and Morgan, 1989).

process. The first level of control, which is already routinely used in the chemical industries, involves sequencing operations, such as manipulating valves or starting or stopping pumps, instrument recalibration, on-line maintenance and fail-safe shut-down procedures. In most of these operations the time base is at least in the order of minutes, so that high-speed manipulations are not vital. Two applications in fermentation processes are sterilization cycles and medium batching.

The next level of computer control involves process control of temperature, pH, foam control, etc. where the sensors are directly interfaced to a computer (Direct Digital Control (DDC); Fig. 8.25). When this is done separate controller units are not needed. The computer program determines the set point values and the control algorithms, such as PID, are part of the computer software package. Better control is possible as the control algorithms are mathematically stored functions rather than electrical functions. This procedure allows for greater flexibility and more precise representation of a process control policy. The system is not very expensive as separate electronic controllers are no longer needed, but computer failure can cause major problems unless there is some manual back-up facility.

The alternative approach is to use a computer in a purely supervisory role. All control functions are performed by an electronic controller using a system illustrated in Fig. 8.26 where the linked computer only logs data from sensors and sends signals to alter set points when instructed by a computer program or manually. This system is known as Supervisory Set-Point Control (SSC) or Digital Set-Point Control (DSC). When SSC is used, the modes of control are limited to proportional, integral and derivative because the direct control of the fermenter is by an electronic controller. However, in the event of computer failure the process controller can be operated independently.

Whiteside and Morgan (1989) have discussed some of the relative merits of DDC and SSC systems and given case histories of the installation and operation of both systems.

The most advanced level of control is concerned with process optimization. This will involve understanding a process, being able to monitor what is happening and being able to control it to achieve and maintain optimum conditions. Firstly, there is a need for suitable on-line sensors to monitor the process continuously. A number are now available for dissolved oxygen, dissolved carbon dioxide, pH, temperature, biomass (the bug meter, NADH fluorescence, near infra-red spec-

troscopy) and some metabolites (mass spectroscopy and near infra-red spectroscopy). All these sensors have been discussed earlier in this chapter. Secondly, it is important to develop a mathematical model that adequately describes the dynamic behaviour of a process. Shimizu (1993) has stressed the vital role which these models play in optimization and reviewed the use of this approach in batch, fed-batch and continuous processes for biomass and metabolites. This approach with appropriate on-line sensors and suitable model programs has been used to optimize bakers' yeast production (Ramirez *et al.*, 1981; Shi *et al.*, 1989), an industrial antibiotic process and lactic acid production (Shi *et al.*, 1989).

Although much progress has been made in the ability to control a process, few sensors are yet available to monitor on-line for many metabolites or other parameters in a fermentation broth thus delaying or making a fast response difficult for on-line control action. Also, it is possible that not all the important parameters in a process have been identified and the mathematical model derived to describe a process may be inadequate. Because of these limitations, an artificial neural network may be used to achieve better control (Karim and Rivera, 1992). These are highly interconnected networks of non-linear processing units arranged in layers with adjustable connecting strengths (weights).

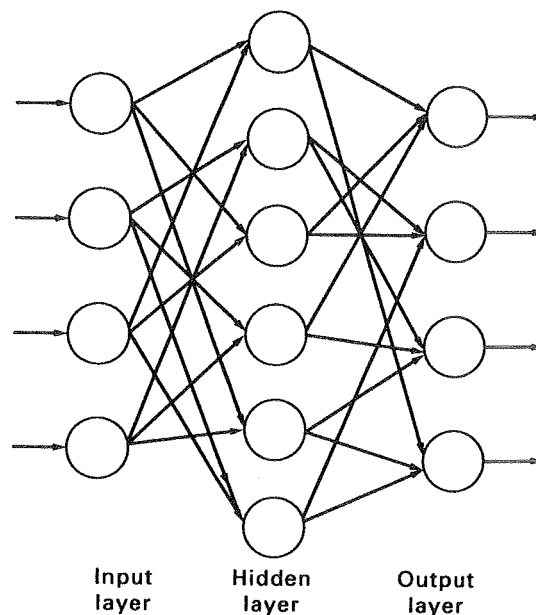


Fig. 8.27. Two-layer neural network (not all the possible interconnections are shown).

In simpler neural networks there is one input layer, one hidden layer and one output layer (Fig. 8.27). Unlike recognized knowledge-based systems, neural networks do not need information in the form of a series of rules, but learn from process examples from which they derive their own rules. This makes it possible to deal with non-linear systems and approximate or limited data.

When training a neural network the aim is to adjust the strengths of the interconnections (neurons) so that a set of inputs produces a desired set of outputs. The inputs may be process variables such as temperature, pH, flow rates, pressure and other direct or indirect measurements which give information about the state of the process. The process outputs obtained (biomass, product, etc.) produce the teacher signal(s) which trains the network. The difference between the desired output and the value predicted by the network is the prediction error. Adjustments are made to minimize the total prediction error by modifying the interconnection strengths until no further decrease in error is achieved. Commercial computer packages are now available to help to determine which of the input variables to use for training and to determine the optimum number of interconnections and hidden layers (Glassey *et al.*, 1994). Readers requiring more detail of the theory of neural networks should consult Karim and Rivera (1992).

This method of control is still at an early stage of development, but it has already been used in a case study on ethanol production by *Zymomonas mobilis* (Karim and Rivera, 1992), in real-time variable estimation and control of a glucoamylase fermentation (Linko and Zhu, 1992) and recombinant *Escherichia coli* fermentations (Glassey *et al.*, 1994).

In industrial systems where a significant amount of on-line and off-line process data may be available, but there are tight time restraints imposed on process optimization, the potential for developing a relatively accurate neural network model within short time scales becomes very attractive (Glassey *et al.*, 1994).

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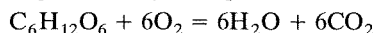
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Aeration and Agitation

INTRODUCTION

THE MAJORITY of fermentation processes are aerobic and, therefore, require the provision of oxygen. If the stoichiometry of respiration is considered, then the oxidation of glucose may be represented as:



Thus, 192 grams of oxygen are required for the complete oxidation of 180 grams of glucose. However, both components must be in solution before they are available to a micro-organism and oxygen is approximately 6000 times less soluble in water than is glucose (a fermentation medium saturated with oxygen contains approximately 7.6 mg dm^{-3} of oxygen at 30°C). Thus, it is not possible to provide a microbial culture with all the oxygen it will need for the complete oxidation of the glucose (or any other carbon source) in one addition. Therefore, a microbial culture must be supplied with oxygen during growth at a rate sufficient to satisfy the organisms' demand.

The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth. However, the productivity of many fermentations is limited by oxygen availability and, therefore, it is important to consider the factors which affect a fermenter's efficiency in supplying microbial cells with oxygen. This chapter considers the requirement for oxygen in fermentation processes, the quantification of oxygen transfer and the factors which will influence the rate of oxygen transfer into solution.

THE OXYGEN REQUIREMENTS OF INDUSTRIAL FERMENTATIONS

Although a consideration of the stoichiometry of

respiration gives an appreciation of the problem of oxygen supply, it gives no indication of an organism's true oxygen demand as it does not take into account the carbon that is converted into biomass and products. A number of workers have considered the overall stoichiometry of the conversion of oxygen, a source of carbon and a source of nitrogen into biomass and have used such relationships to predict the oxygen demand of a fermentation. A selection of such equations is shown in Table 9.1. From these determinations it may be seen that a culture's demand for oxygen is very much dependent on the source of carbon in the medium. Thus, the more reduced the carbon source, the greater will be the oxygen demand. From Darlington's and Johnson's equations (Table 9.1) it may be seen that the production of 100 grams of biomass from hydrocarbon requires approximately three times the amount of oxygen to produce the same amount of biomass from carbohydrate. This point is also illustrated in Table 9.2. However, it must be remembered that the high carbon content of hydrocarbon substrates means that high yield factors (g biomass g^{-1} substrate consumed) are obtained and the decision to use such substrates is based on the balance between the advantage of high biomass yield and the disadvantage of high oxygen demand and heat generation. These points are discussed in more detail in Chapter 4.

Darlington's, Johnson's and Mateles' equations only include biomass production and do not consider product formation, whereas Cooney's and Righelato's equations consider product formation. Ryu and Hospodka (1980) used Righelato's approach to calculate that the production of 1 g penicillin consumes 2.2 g of oxygen.

However, it is inadequate to base the provision of oxygen for a fermentation simply on an estimation of overall demand, because the metabolism of the culture is affected by the concentration of dissolved oxygen in the broth. The effect of dissolved oxygen concentration

TABLE 9.1. Stoichiometric equations describing oxygen demand in a fermentation

Equation	Terms used	Reference
$6.67\text{CH}_2\text{O} + 2.1\text{O}_2 = \text{C}_{3.92}\text{H}_{6.5}\text{O}_{1.94} + 2.75\text{CO}_2 + 3.42\text{H}_2\text{O}$	$\text{C}_{3.92}\text{H}_{6.5}\text{O}_{1.94}$ is 100 g (dry weight) of yeast cells; CH_2O is carbohydrate	Darlington (1964)
$7.14\text{CH}_2 + 6.135\text{O}_2 = \text{C}_{3.92}\text{H}_{6.5}\text{O}_{1.94} + 3.22\text{CO}_2 + 3.89\text{H}_2\text{O}$	CH_2 is hydrocarbon	Darlington (1964)
$(A/Y) - B = C$	A = Amount of oxygen for combustion of 1 g of substrate to CO_2 , H_2O and NH_3 , if nitrogen is present in the substrate B = Amount of oxygen required for the combustion of 1 g cells to CO_2 , H_2O and NH_3 Y = Cell yield (g cells g^{-1} substrate) C = g oxygen consumed for the production of 1 g of cells	Johnson (1964)
$(dx/dt)/Y + mx + p = (d\text{CO}_2/dt) = -(d\text{O}_2/dt)RQ$	x = biomass concentration t = time Y = g biomass g^{-1} carbon substrate m = maintenance p = allowance for antibiotic production	Righelato <i>et al.</i> (1968)
$Y_{O/P} = (0.53/Y_{P/G}) - (0.6X/P) - 0.43$	$Y_{O/P}$ = g oxygen consumed g^{-1} glucose $Y_{P/G}$ = g sodium penicillin G produced g^{-1} glucose X = g cells (dry weight) produced P = g sodium penicillin G produced	Cooney (1979)
$Y_O = \{(32C + 8H - 16O)/YM\} - 1.58$	Y_O = g oxygen consumed g^{-1} cells produced Y = Cell yield (g cells g^{-1} substrate) M = Molecular weight of the carbon source C, H and O = Number of atoms of carbon, hydrogen and oxygen per molecule of carbon source	Mateles (1971)

on the specific oxygen uptake rate (Q_{O_2} , mmoles of oxygen consumed per gram dry weight of cells per hour) has been shown to be of the Michaelis-Menten type, as shown in Fig. 9.1.

From Fig. 9.1 it may be seen that the specific oxygen uptake rate increases with increase in the dissolved oxygen concentration up to a certain point (referred to as C_{crit}) above which no further increase in oxygen uptake rate occurs. Some examples of the critical oxygen levels for a range of micro-organisms are given in

Table 9.3. Thus, maximum biomass production may be achieved by satisfying the organism's maximum specific oxygen demand by maintaining the dissolved oxygen concentration greater than the critical level. If the dissolved oxygen concentration were to fall below the critical level then the cells may be metabolically disturbed. However, it must be remembered that it is frequently the objective of the fermentation technologist to produce a product of the micro-organism rather than the organism itself and that metabolic

TABLE 9.2. Oxygen requirements of a range of micro-organisms grown on a range of substrates (After Mateles, 1979)

Substrate	Organism	Oxygen requirement (g O_2 g^{-1} dry wt.)	Reference
Glucose	<i>Escherichia coli</i>	0.4	Schulze and Lipe (1964)
Methanol	<i>Pseudomonas C</i>	1.2	Goldberg <i>et al.</i> (1976)
Octane	<i>Pseudomonas sp.</i>	1.7	Wodzinski and Johnson (1968)

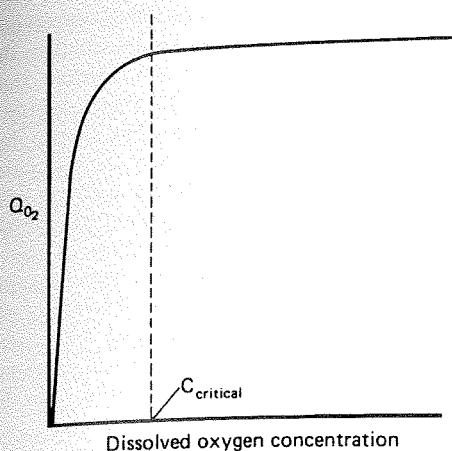


FIG. 9.1. The effect of dissolved oxygen concentration on the Q_{O_2} of a micro-organism.

disturbance of the cell by oxygen starvation may be advantageous to the formation of certain products. Equally, provision of a dissolved oxygen concentration far greater than the critical level may have no influence on biomass production, but may stimulate product formation. Thus, the aeration conditions necessary for the optimum production of a product may be different from those favouring biomass production.

Hirose and Shibai's (1980) investigations of amino acid biosynthesis by *Brevibacterium flavum* provide an excellent example of the effects of the dissolved oxygen concentration on the production of a range of closely related metabolites. These workers demonstrated the critical dissolved oxygen concentration for *B. flavum* to be 0.01 mg dm^{-3} and considered the extent of oxygen supply to the culture in terms of the degree of 'oxygen satisfaction', that is the respiratory rate of the culture expressed as a fraction of the maximum respiratory rate. Thus, a value of oxygen satisfaction below unity implied that the dissolved oxygen concentration was below the critical level. The effect of the degree of

oxygen satisfaction on the production of a range of amino acids is shown in Fig. 9.2. From Fig. 9.2 it may be seen that the production of members of the glutamate and aspartate families of amino acids was affected detrimentally by levels of oxygen satisfaction below 1.0, whereas optimum production of phenylalanine, valine and leucine occurred at oxygen satisfaction levels of 0.55, 0.60 and 0.85, respectively. The biosynthetic routes of the amino acids are shown in Fig. 9.3, from which it may be seen that the glutamate and aspartate families are all produced from tricarboxylic acid (TCA) cycle intermediates, whereas phenylalanine, valine and leucine are produced from the glycolysis intermediates, pyruvate and phosphoenol pyruvate. Oxygen excess should give rise to abundant TCA cycle intermediates, whereas oxygen limitation should result in less glucose being oxidized via the TCA cycle, allowing more intermediates to be available for phenylalanine, valine and leucine biosynthesis. Thus, some degree of metabolic disruption results in greater production of pyruvate derived amino acids.

An example of the effect of dissolved oxygen on secondary metabolism is provided by Zhou *et al.*'s (1992) work on cephalosporin C synthesis by *Cephalosporium acremonium*. These workers demonstrated that the critical oxygen concentration for cephalosporin C synthesis during the production phase was 20% saturation. At dissolved oxygen concentrations below 20% cephalosporin C concentration declined and penicillin N increased. The biosynthetic pathway to cephalosporin C is shown in Fig. 9.4, from which it may be seen that there are three oxygen-consuming steps in the pathway:

- (i) Cyclization of the tripeptide, α -amino-adipyl-cysteinyl-valine into isopenicillin N.
- (ii) The ring expansion of penicillin N into deacetoxycephalosporin C (DAOC).
- (iii) The hydroxylation of DAOC to give deacetylcephalosporin C.

TABLE 9.3. Critical dissolved oxygen concentrations for a range of micro-organisms (Riviere, 1977)

Organism	Temperature	Critical dissolved oxygen concentration (mmoles dm^{-3})
<i>Azotobacter</i> sp.	30	0.018
<i>Escherichia coli</i>	37	0.008
<i>Saccharomyces</i> sp.	30	0.004
<i>Penicillium chrysogenum</i>	24	0.022

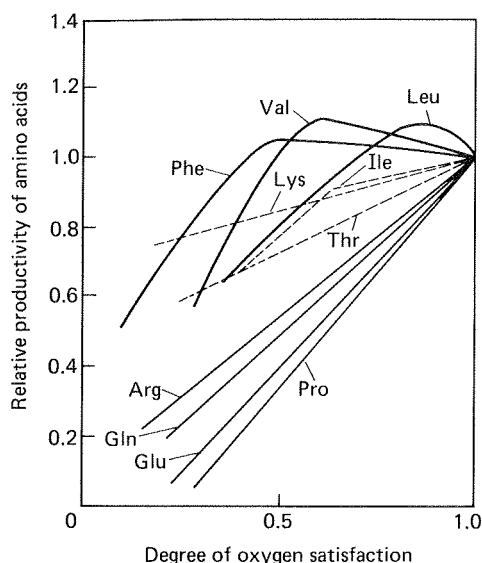


FIG. 9.2. The effect of dissolved oxygen on the production of amino acids by *Brevibacterium flavum* (Hirose and Shibai, 1980).

DAOC did not accumulate at low oxygen concentrations and, thus, it appears that the most oxygen sensitive step in the pathway is the ring expansion enzyme (expandase) resulting in the accumulation of penicillin N under oxygen limitation.

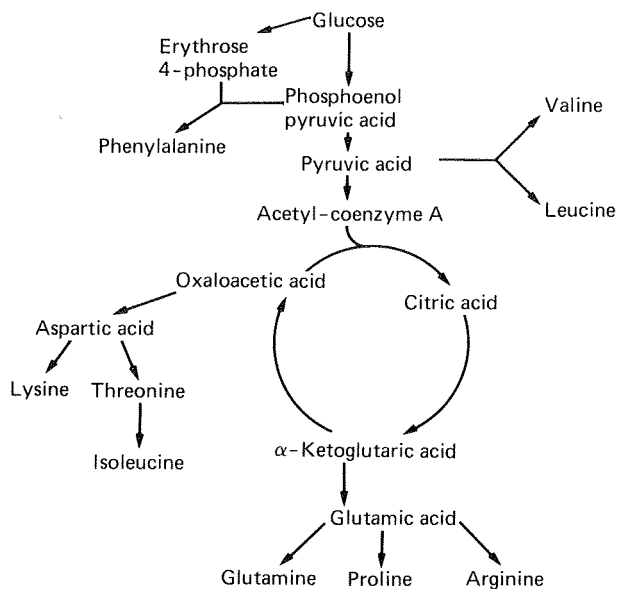


FIG. 9.3. The biosynthetic routes to the amino acids phenylalanine, valine, leucine, lysine, threonine, *l*-leucine, glutamic acid, proline, glutamine and arginine in *B. flavum*.

The requirement for a high dissolved oxygen concentration by many fermentations has resulted in the development of process techniques to ensure that the fermentation does not exceed the oxygen-supply capabilities of the fermentation vessel. The oxygen demand of a fermentation largely depends on the concentration of the biomass and its respiratory activity, which is related to the growth rate. By limiting the initial concentration of the medium, the biomass in the vessel may be kept at a reasonable level and by supplying some nutrient component as a feed, the rate of growth, and hence the respiratory rate, may be controlled. These techniques of medium design and nutrient feed are discussed in Chapters 2 and 4 and later in this chapter.

OXYGEN SUPPLY

Oxygen is normally supplied to microbial cultures in the form of air, this being the cheapest available source

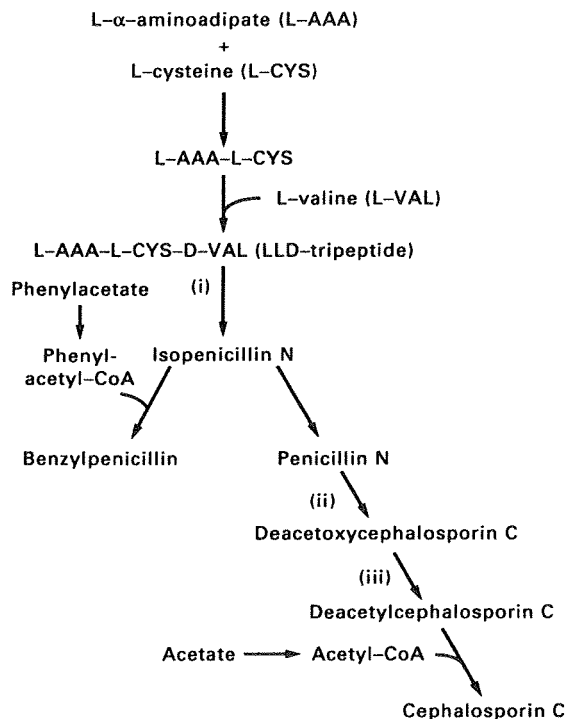


FIG. 9.4. The biosynthesis of cephalosporin C, indicating the oxygen consuming steps:

- (i) isopenicillin-N-synthase,
- (ii) deacetoxycephalosporin C synthase (commonly called expandase),
- (iii) deacetyl cephalosporin C synthase (commonly called hydroxylase).

of the gas. The method for provision of a culture with a supply of air varies with the scale of the process:

- (i) Laboratory-scale cultures may be aerated by means of the shake-flask technique where the culture (50 to 100 cm³) is grown in a conical flask (250 to 500 cm³) shaken on a platform contained in a controlled environment chamber.
- (ii) Pilot- and industrial-scale fermentations are normally carried out in stirred, aerated vessels, termed fermenters, of the type described in Chapter 7. However, it is often advantageous to culture relatively small volumes (1 dm³) in a stirred, aerated vessel as this enables the cultural conditions to be better monitored and controlled, and facilitates the addition of supplements and the removal of samples. Some fermenters are so designed that adequate oxygen transfer is obtained without agitation and the design of these systems (termed bubble columns and air-lift fermenters) is also discussed in Chapter 7.

Bartholomew *et al.* (1950) represented the transfer of oxygen from air to the cell, during a fermentation, as occurring in a number of steps:

- (i) The transfer of oxygen from an air bubble into solution.
- (ii) The transfer of the dissolved oxygen through the fermentation medium to the microbial cell.
- (iii) The uptake of the dissolved oxygen by the cell.

These workers demonstrated that the limiting step in the transfer of oxygen from air to the cell in a *Streptomyces griseus* fermentation was the transfer of oxygen into solution. These findings have been shown to be correct for non-viscous fermentations but it has been demonstrated that transfer may be limited by either of the other two stages in certain highly viscous fermentations. The difficulties inherent in such fermentations are discussed later in this chapter.

The rate of oxygen transfer from air bubble to the liquid phase may be described by the equation:

$$dC_L/dt = K_L a (C^* - C_L) \quad (9.1)$$

where C_L is the concentration of dissolved oxygen in the fermentation broth (mmoles dm⁻³),
 t is time (hours),

dC_L/dt is the change in oxygen concentration over a time period, i.e. the oxygen-transfer rate (mmoles O₂ dm⁻³ h⁻¹),
 K_L is the mass transfer coefficient (cm h⁻¹),
 a is the gas/liquid interface area per liquid volume (cm² cm⁻³),
 C^* is the saturated dissolved oxygen concentration (mmoles dm⁻³).

K_L may be considered as the sum of the reciprocals of the resistances to the transfer of oxygen from gas to liquid and $(C^* - C_L)$ may be considered as the 'driving force' across the resistances. It is extremely difficult to measure both K_L and ' a ' in a fermentation and, therefore, the two terms are generally combined in the term $K_L a$, the volumetric mass-transfer coefficient, the units of which are reciprocal time (h⁻¹). The volumetric mass-transfer coefficient is used as a measure of the aeration capacity of a fermenter. The larger the $K_L a$, the higher the aeration capacity of the system. The $K_L a$ value will depend upon the design and operating conditions of the fermenter and will be affected by such variables as aeration rate, agitation rate and impeller design. These variables affect ' K_L ' by reducing the resistances to transfer and affect ' a ' by changing the number, size and residence time of air bubbles. It is convenient to use $K_L a$ as a yardstick of fermenter performance because, unlike the oxygen-transfer rate, it is unaffected by dissolved oxygen concentration. However, the oxygen transfer rate is the critical criterion in a fermentation and, as may be seen from equation 9.1, it is affected by both $K_L a$ and dissolved oxygen concentration. The dissolved oxygen concentration reflects the balance between the supply of dissolved oxygen by the fermenter and the oxygen demand of the organism. If the $K_L a$ of the fermenter is such that the oxygen demand of the organism cannot be met, the dissolved oxygen concentration will decrease below the critical level (C_{crit}). If the $K_L a$ is such that the oxygen demand of the organism can be easily met the dissolved oxygen concentration will be greater than C_{crit} and may be as high as 70 to 80% of the saturation level. Thus, the $K_L a$ of the fermenter must be such that the optimum oxygen concentration for product formation can be maintained in solution throughout the fermentation.

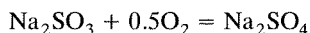
DETERMINATION OF $K_L a$ VALUES

The determination of the $K_L a$ of a fermenter is essential in order to establish its aeration efficiency

and to quantify the effects of operating variables on the provision of oxygen. This section considers the merits and limitations of the methods available for the determination of K_La values. It is important to remember at this stage that dissolved oxygen is usually monitored using a dissolved oxygen electrode (see Chapter 8) which records dissolved oxygen activity or dissolved oxygen tension (DOT) whilst the equations describing oxygen transfer are based on dissolved oxygen concentration. The solubility of oxygen is affected by dissolved solutes so that pure water and a fermentation medium saturated with oxygen would have different dissolved oxygen concentrations yet have the same DOT, i.e. an oxygen electrode would record 100% for both. Thus, to translate DOT into concentration the solubility of oxygen in the fermentation medium must be known and this can present difficulties.

The sulphite oxidation technique

Cooper *et al.* (1944) were the first to describe the determination of oxygen-transfer rates in aerated vessels by the oxidation of sodium sulphite solution. This technique does not require the measurement of dissolved oxygen concentrations but relies on the rate of conversion of a 0.5 M solution of sodium sulphite to sodium sulphate in the presence of a copper or cobalt catalyst:



The rate of reaction is such that as oxygen enters solution it is immediately consumed in the oxidation of sulphite, so that the sulphite oxidation rate is equivalent to the oxygen-transfer rate. The dissolved oxygen concentration, for all practical purposes, will be zero and the K_La may then be calculated from the equation:

$$\text{OTR} = K_La \cdot C^* \quad (9.2)$$

(where OTR is the oxygen transfer rate).

The procedure is carried out as follows: the fermenter is batched with a 0.5 M solution of sodium sulphite containing 10^{-3} M Cu^{2+} ions and aerated and agitated at fixed rates; samples are removed at set time intervals (depending on the aeration and agitation rates) and added to excess iodine solution which reacts with the unconsumed sulphite, the level of which may be determined by a back titration with standard sodium thiosulphate solution. The volumes of the thiosulphate titrations are plotted against sample time and the oxygen transfer rate may be calculated from the slope of the graph.

The sulphite oxidation method has the advantage of simplicity and, also, the technique involves sampling the bulk liquid in the fermenter and, therefore, removes some of the problems of conditions varying through the volume of the vessel. However, the method is time consuming (one determination taking up to 3 hours, depending on the aeration and agitation rates) and is notoriously inaccurate. Bell and Gallo (1971) demonstrated that minor amounts of surface-active contaminants (such as amino acids, proteins, fatty acids, esters, lipids, etc.) could have a major effect on the accuracy of the technique and apparent differences in aeration efficiency between vessels could be due to differences in the degree of contamination. Also, the rheology of a sodium sulphite solution is completely different from that of a fermentation broth, especially a mycelial one so that it is impossible to relate the results of sodium sulphite determinations to real fermentations. To quote Van't Riet and Tramper (1991) "It can safely be said that the application of this method should be strongly discouraged".

Gassing-out techniques

The estimation of the K_La of a fermentation system by gassing-out techniques depends upon monitoring the increase in dissolved oxygen concentration of a solution during aeration and agitation. The oxygen transfer rate will decrease during the period of aeration as C_L approaches C^* due to the decline in the driving force ($C^* - C_L$). The oxygen transfer rate, at any one time, will be equal to the slope of the tangent to the curve of values of dissolved oxygen concentration against time of aeration, as shown in Fig. 9.5.

To monitor the increase in dissolved oxygen over an adequate range it is necessary first to decrease the oxygen level to a low value. Two methods have been employed to achieve this lowering of the dissolved oxygen concentration — the static method and the dynamic method.

THE STATIC METHOD OF GASSING OUT

In this technique, first described by Wise (1951), the oxygen concentration of the solution is lowered by gassing the liquid out with nitrogen gas, so that the solution is 'scrubbed' free of oxygen. The deoxygenated liquid is then aerated and agitated and the increase in dissolved oxygen monitored using some form of dissolved oxygen probe. The increase in dissolved oxygen

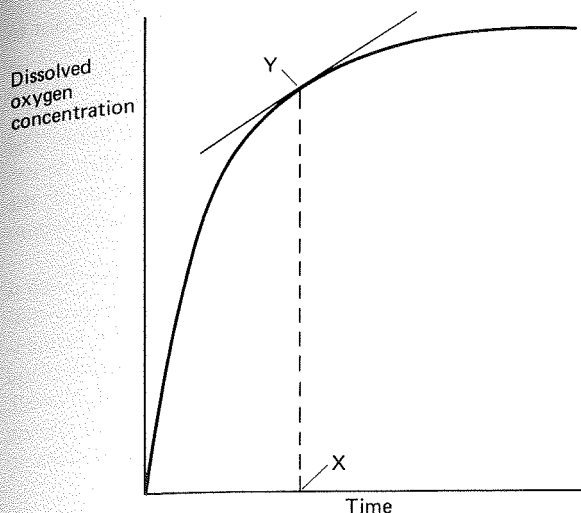


FIG. 9.5. The increase in dissolved oxygen concentration of a solution over a period of aeration. The oxygen transfer rate at time X is equal to the slope of the tangent at point Y.

concentration has already been described by equation (9.1), i.e.:

$$dC_L/dt = K_L a(C^* - C_L)$$

and depicted in Fig. 9.5. Integration of equation (9.1) yields:

$$\ln(C^* - C_L) = -K_L a t. \quad (9.3)$$

Thus, a plot of $\ln(C^* - C_L)$ against time will yield a straight line of slope $-K_L a$, as shown in Fig 9.6. This technique has the advantage over the sulphite oxidation method in that it is very rapid (normally taking up to 15 minutes) and may utilize the fermentation medium, to which may be added dead cells or mycelium at a concentration equal to that produced during the fermentation. However, employing the fermentation medium with, or without, killed biomass necessitates the use of a membrane-type electrode, the response time of which may be inadequate to reflect the true change in the rate of oxygenation over a short period of time. The probe response time (T_p) is defined as the time needed to record 63% of a stepwise change and this should be much smaller than the mass transfer response time of the system ($1/K_L a$). According to Van't Riet (1979), the use of commercially available electrodes, with a response time of 2 to 3 seconds, should enable a $K_L a$ of up to 360 h^{-1} to be measured with little loss of accuracy. However, for estimations of higher $K_L a$ values it would be necessary to incorporate a correction factor into the calculation, as discussed by Taguchi and Humphrey (1966), Heineken (1970, 1971)

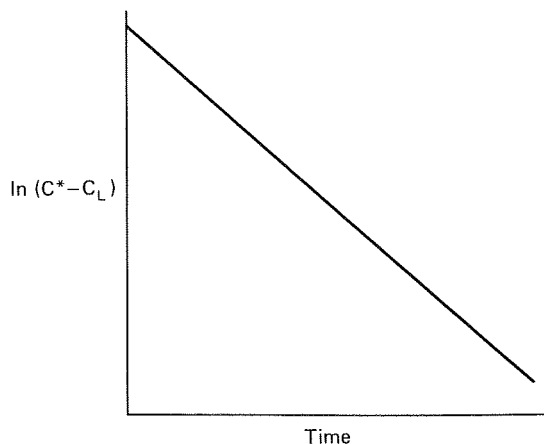


FIG. 9.6. A plot of the $\ln(C^* - C_L)$ against time of aeration, the slope of which equals $-K_L a$.

and Wernau and Wilke (1973). It is not necessary to know the oxygen solubility in the medium because DOT values may be used directly in order to calculate the rates, i.e. C^* is taken as 100%.

Whilst the method is acceptable for small scale vessels, there are severe limitations to its use on large scale fermenters which have high gas residence times. When the air supply to such a vessel is resumed after deoxygenation with nitrogen, the oxygen concentration in the gas phase may change with time as the nitrogen is replaced with air. Thus, C^* will no longer be constant. Although correction factors have been derived to compensate for this phenomenon, Van't Riet and Tramper (1991) concluded that the method should not be used for vessels over 1-metre high.

THE DYNAMIC METHOD OF GASSING OUT

Taguchi and Humphrey (1966) utilized the respiratory activity of a growing culture in the fermenter to lower the oxygen level prior to aeration. Therefore, the estimation has the advantage of being carried out during a fermentation which should give a more realistic assessment of the fermenter's efficiency. Because of the complex nature of fermentation broths the probe used to monitor the change in dissolved oxygen concentration must be of the membrane-covered type which may necessitate the use of the response-correction factors referred to previously. The procedure involves stopping the supply of air to the fermentation which results in a linear decline in the dissolved oxygen concentration due to the respiration of the culture, as shown in Fig. 9.7. The slope of the line AB in Fig. 9.7 is

a measure of the respiration rate of the culture. At point B the aeration is resumed and the dissolved oxygen concentration increases until it reaches concentration X. Over the period, BC, the observed increase in dissolved oxygen concentration is the difference between the transfer of oxygen into solution and the uptake of oxygen by the respiring culture as expressed by the equation:

$$dC_L/dt = K_L a (C^* - C_L) - xQ_{O_2} \quad (9.4)$$

where x is the concentration of biomass and

Q_{O_2} is the specific respiration rate (mmoles of oxygen g^{-1} biomass h^{-1}).

The term xQ_{O_2} is given by the slope of the line AB in Fig. 9.7. Equation (9.4) may be rearranged as:

$$C_L = -1/K_L a \{ (dC_L/dt) + xQ_{O_2} \} + C^* \quad (9.5)$$

Thus, from equation (9.5), a plot of C_L versus $dC_L/dt + xQ_{O_2}$ will yield a straight line, the slope of which will equal $-1/K_L a$, as shown in Fig. 9.8. This technique is convenient in that the equations may be applied using DOT rather than concentration because it is the rates of transfer and uptake that are being monitored so that the percentage saturation readings generated by the electrode may be used directly.

The dynamic gassing-out method has the advantage over the previous methods of determining the $K_L a$ during an actual fermentation and may be used to determine $K_L a$ values at different stages in the process. The technique is also rapid and only requires the use of a dissolved-oxygen probe, of the membrane type. A major limitation in the operation of the technique is the range over which the increase in dissolved oxygen

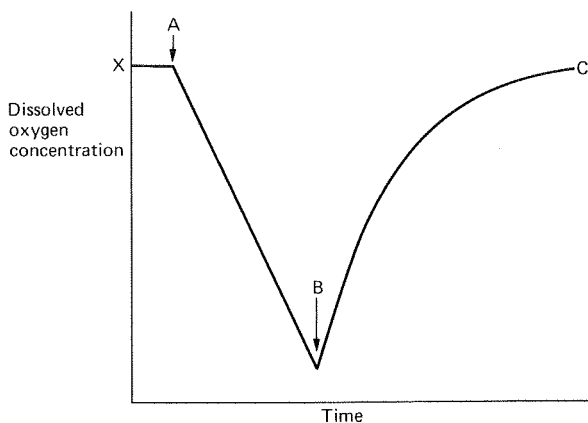


FIG. 9.7. Dynamic gassing out for the determination of $K_L a$ values. Aeration was terminated at point A and recommenced at point B.

concentration may be measured. It is important not to allow the oxygen concentration to drop below C_{crit} during the deoxygenation step as the specific oxygen uptake rate will then be limited and the term xQ_{O_2} would not be constant on resumption of aeration. The occurrence of oxygen-limited conditions during deoxygenation may be detected by the deviation of the decline in oxygen concentration from a linear relationship with time, as shown in Fig. 9.9.

When the oxygen demand of a culture is very high it may be difficult to maintain the dissolved oxygen concentration significantly above C_{crit} during the fermentation so that the range of measurements which could be used in the $K_L a$ determination would be very small. Thus, it may be difficult to apply the technique during a fermentation which has an oxygen demand close to the supply capacity of the fermenter.

Although the difficulty presented by nitrogen degassing does not arise with the dynamic method it, also, is not suitable for use with vessels in excess of one metre high. Van't Riet and Tramper (1991) pointed out that in such vessels the time taken to establish an equilibrium population of air bubbles would be significant and the gas-liquid interface area would change over the aeration period resulting in a considerable underestimate of the $K_L a$ value achievable under normal operating conditions. Both the dynamic and static methods are also unsuitable for measuring $K_L a$ values in viscous systems. This is due to the very small bubbles (< 1 mm diameter) formed in a viscous system which have an extended residence time compared with 'nor-

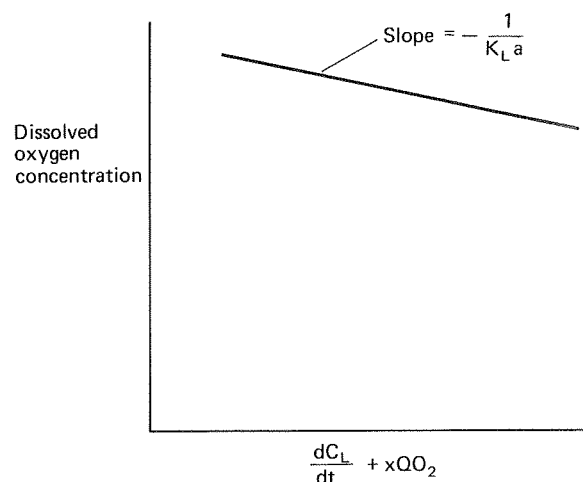


FIG. 9.8. The dynamic method for determination of $K_L a$ values. The information is gleaned from Fig. 9.7, by taking tangents of the curve, BC, at various values of C_L .

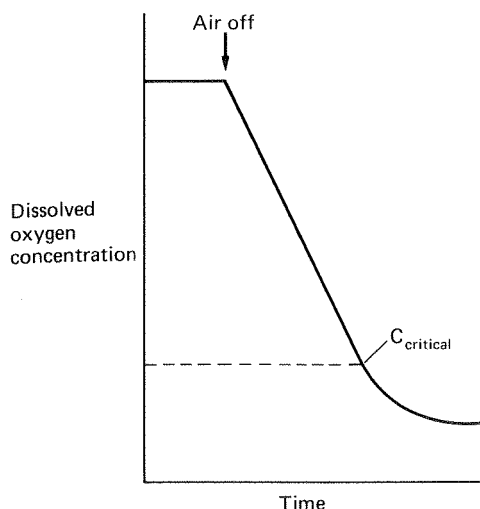


FIG. 9.9. The occurrence of oxygen limitation during the dynamic gassing out of a fermentation.

mal' sized bubbles. Thus, the gassing out techniques are only useful on a small scale with non-viscous systems.

The oxygen-balance technique

The $K_L a$ of a fermenter may be measured during a fermentation by the oxygen balance technique which determines, directly, the amount of oxygen transferred into solution in a set time interval. The procedure involves measuring the following parameters:

- (i) The volume of the broth contained in the vessel, V_L (dm^3).
- (ii) The volumetric air flow rates measured at the air inlet and outlet, Q_i and Q_o , respectively ($\text{dm}^3 \text{ min}^{-1}$).
- (iii) The total pressure measured at the fermenter air inlet and outlet, P_i and P_o , respectively (atm. absolute).
- (iv) The temperature of the gases at the inlet and outlet, T_i and T_o , respectively (K).
- (v) The mole fraction of oxygen measured at the inlet and outlet, y_i and y_o , respectively.

The oxygen transfer rate may then be determined from the following equation (Wang *et al.*, 1979):

$$\text{OTR} = (7.32 \times 10^5 / V_L) (Q_i P_i y_i / T_i - Q_o P_o y_o / T_o) \quad (9.6)$$

where 7.32×10^5 is the conversion factor equalling $(60 \text{ min h}^{-1}) [\text{mole}/22.4 \text{ dm}^3 (\text{STP})] (273 \text{ K}/1 \text{ atm})$.

These measurements require accurate flow meters, pressure gauges and temperature-sensing devices as well as gaseous oxygen analysers (see Chapter 8). The ideal gaseous oxygen analyser is a mass spectrometer analyser which is sufficiently accurate to detect changes of 1 to 2%.

The $K_L a$ may be determined, provided that C_L and C^* are known, from equation (9.1):

$$\text{OTR} = K_L a (C^* - C_L) \text{ or } K_L a = \text{OTR} / (C^* - C_L)$$

C_L may be determined using a membrane-type dissolved-oxygen electrode and in this case the slow response time is not an important factor because a rate of change is not being measured, simply the steady-state oxygen concentration. However, it should be remembered that an electrode simply measures the oxygen tension at one point and it is, therefore, advisable to monitor the oxygen tension at a number of points in the vessel with a number of electrodes and to use an average value. Also, the DOT reading must be converted to concentration, which necessitates knowing the oxygen solubility in the fermentation medium. The value of C^* is frequently taken as that value which is in equilibrium with the oxygen concentration of the gas outlet. Wang *et al.* (1979) claimed that this approach was adequate for small-scale fermenters but on a large scale there may be a considerable difference between the dissolved oxygen concentration in equilibrium with the inlet and outlet gases. Therefore, these workers suggested that the behaviour of the gas in transit in the fermenter would approximate to plug flow conditions and a logarithmic mean value for the dissolved oxygen concentration should be used.

The oxygen-balance technique appears to be the simplest method for the assessment of $K_L a$ and has the advantage of measuring aeration efficiency during a fermentation. The sulphite oxidation and static gassing-out techniques have the disadvantage of being carried out using either a salt solution or an uninoculated, sterile fermentation medium. Although, as Banks (1977) suggests, these techniques are adequate for the comparison of equipment or operating variables, it should not be assumed that the values obtained are those actually operating during a fermentation. This may be the case for bacterial or yeast fermentations where the rheology of the suspended cells in the broth is similar to that in a sterile medium or a salt solution, but it is certainly not true for fungal and streptomycete processes where the rheology is quite different.